

BFS Traversal

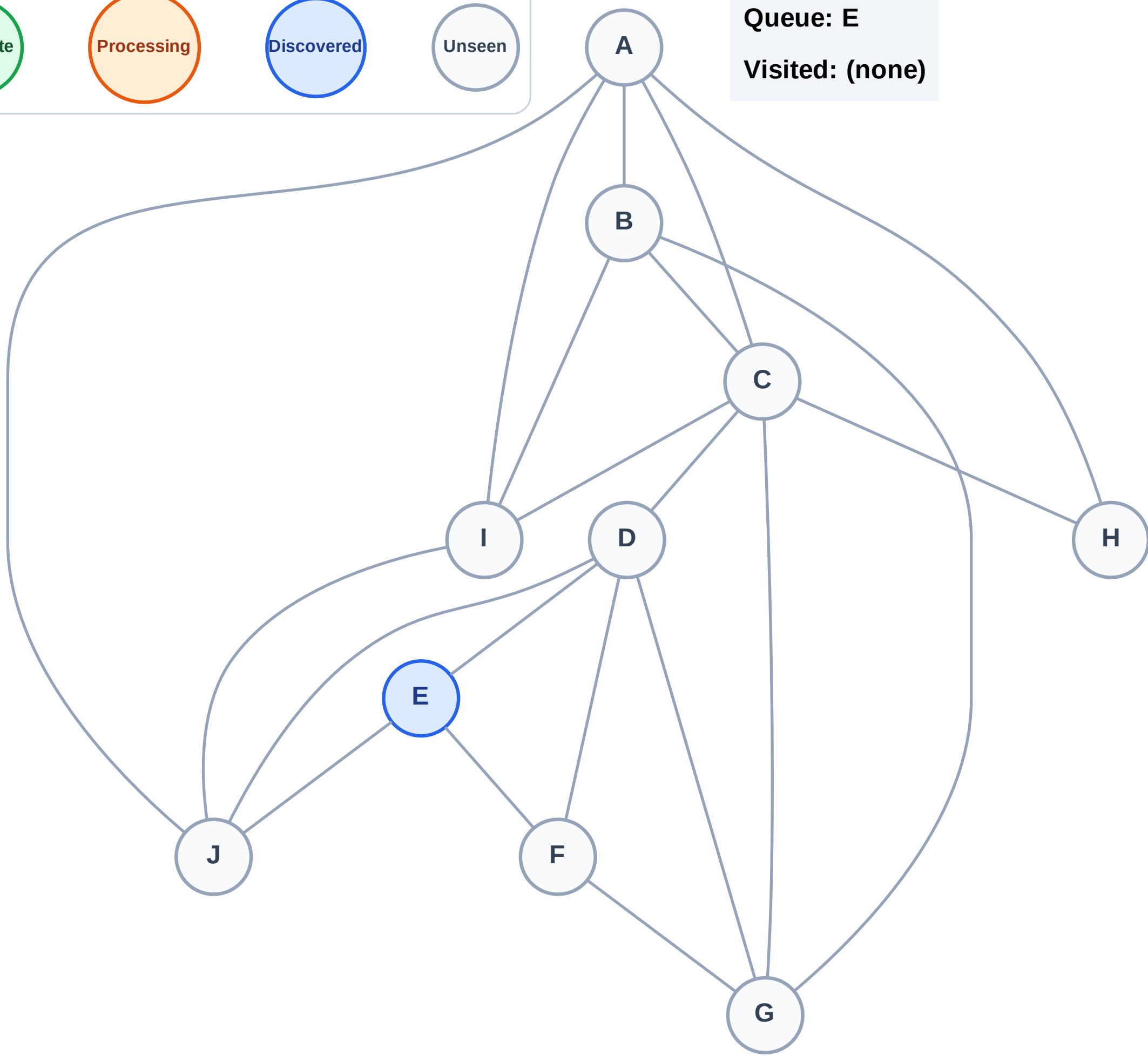
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

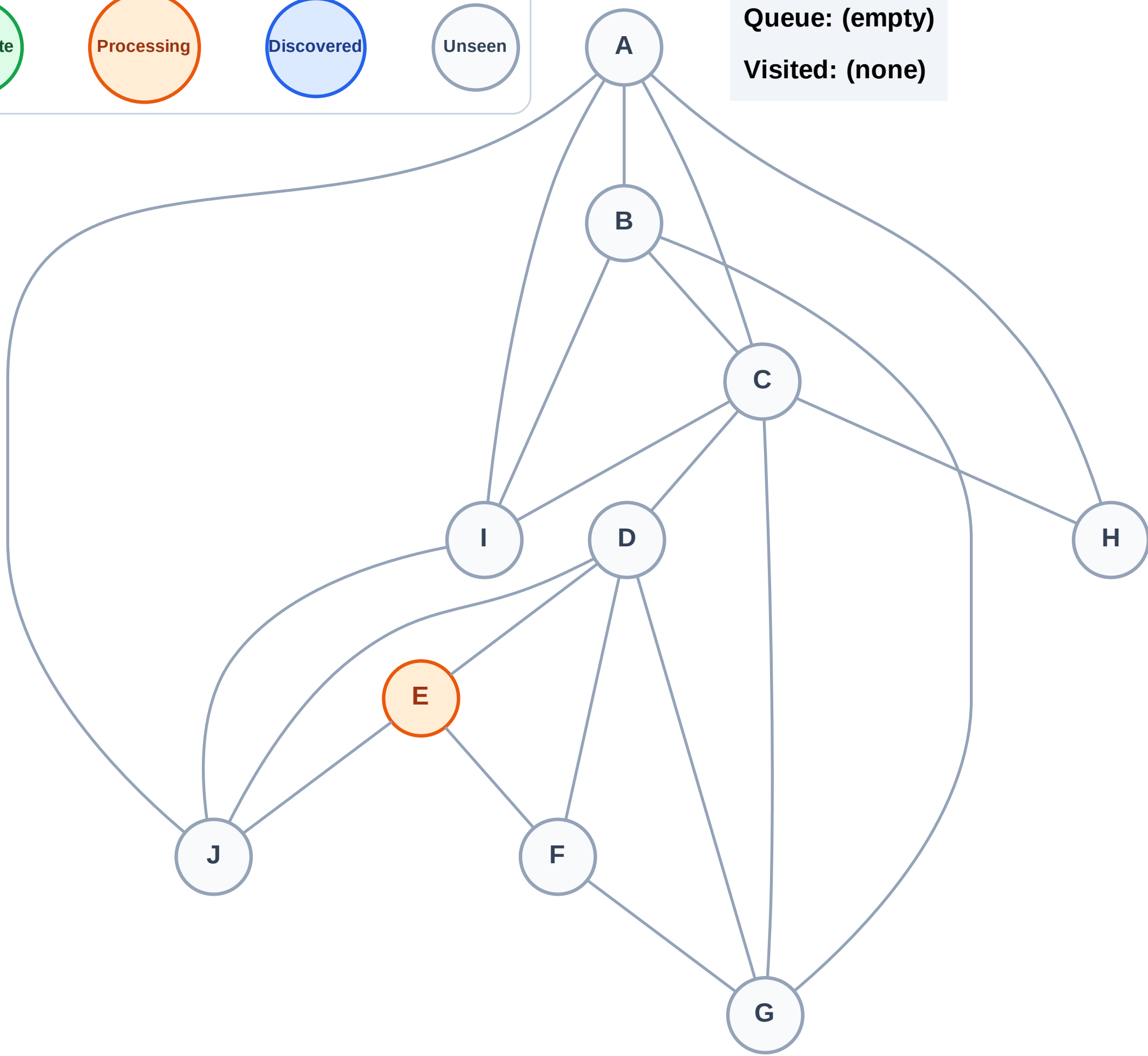
Step 2: Process node E

Legend



Queue: (empty)

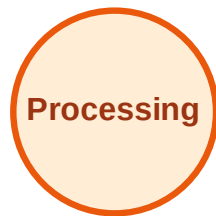
Visited: (none)



BFS Traversal

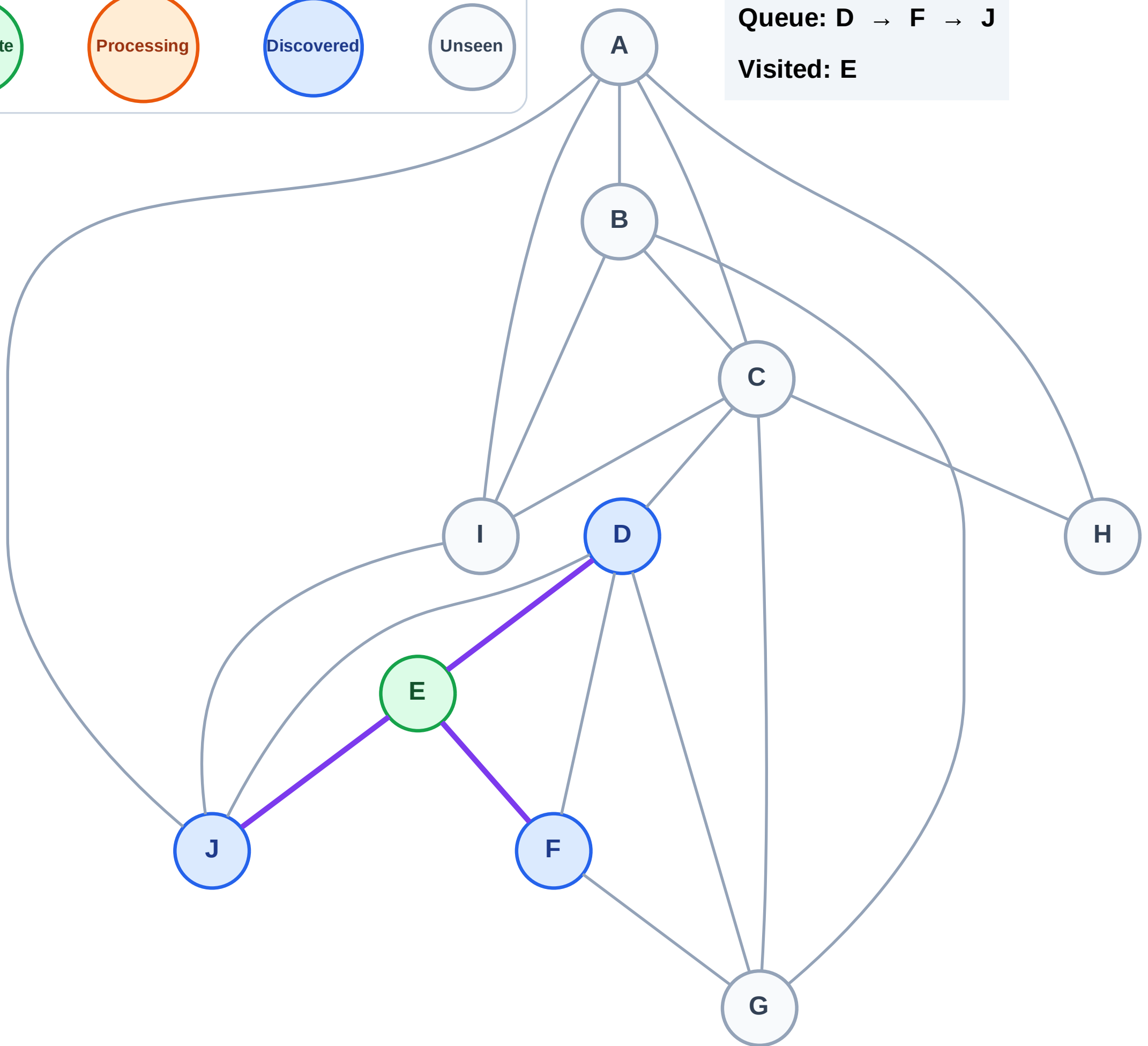
Step 3: Discover D, F, J from E

Legend



Queue: D → F → J

Visited: E



BFS Traversal

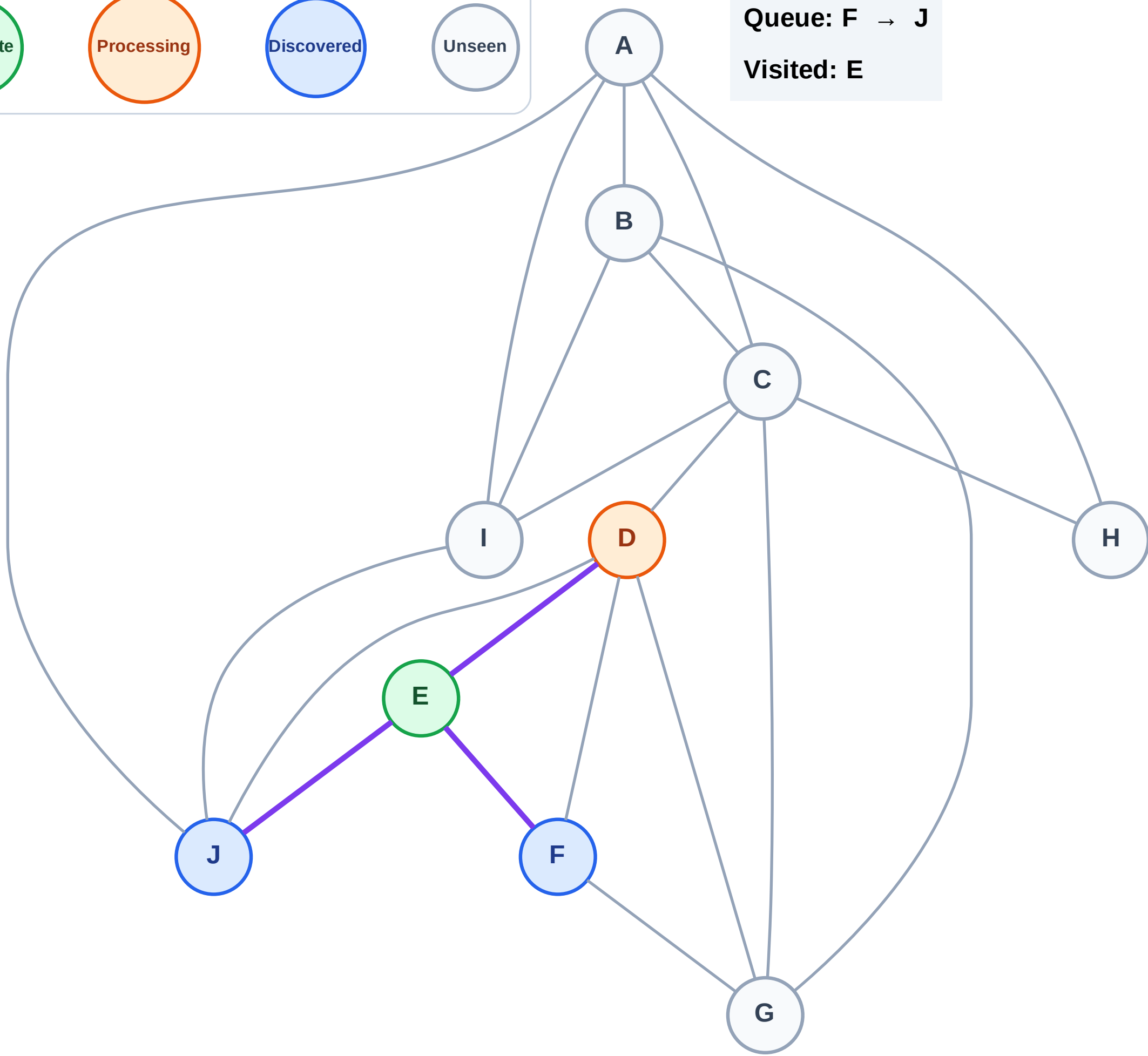
Step 4: Process node D

Legend



Queue: F → J

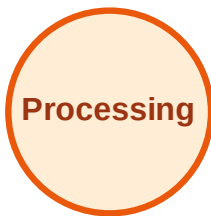
Visited: E



BFS Traversal

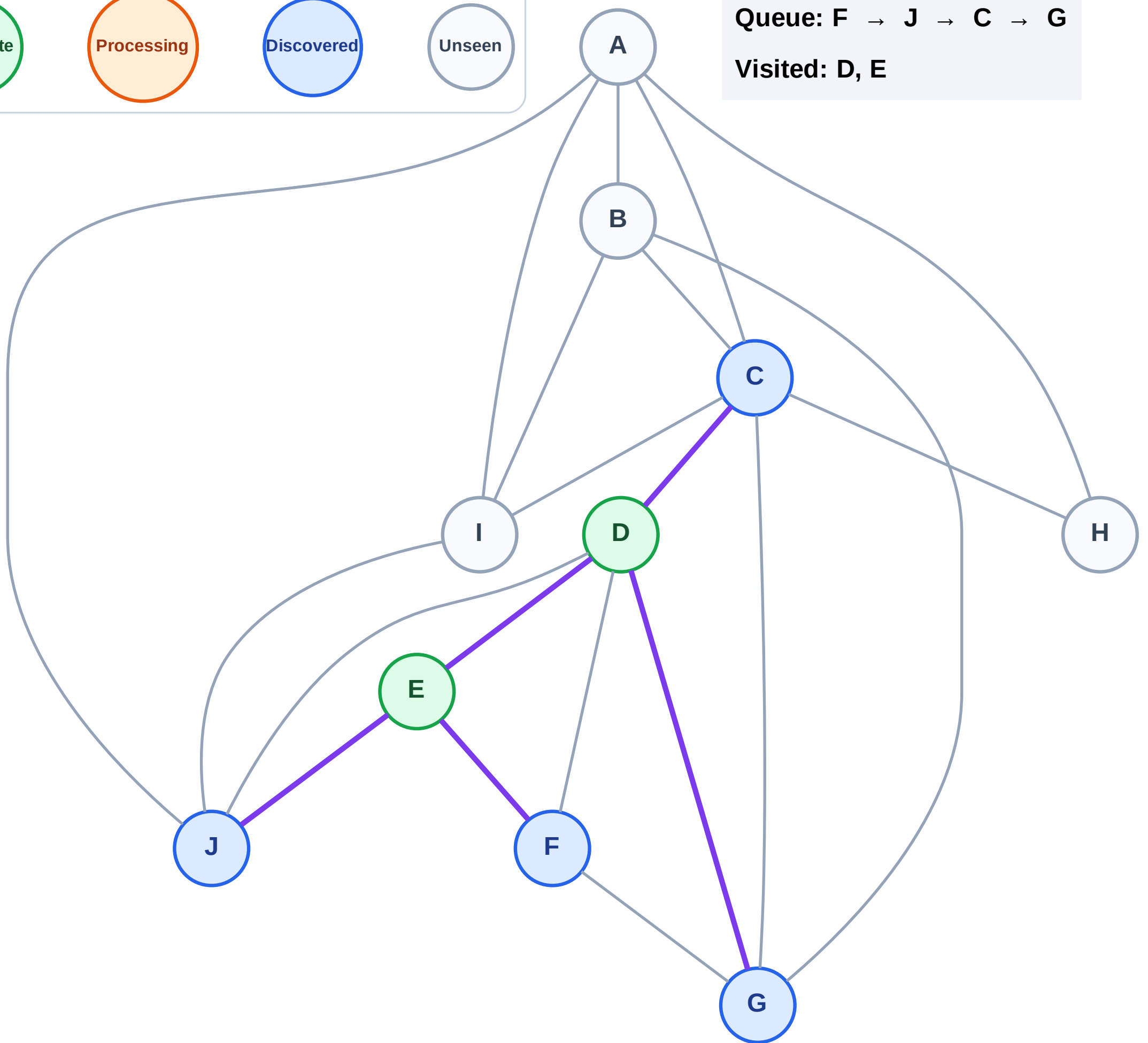
Step 5: Discover C, G from D

Legend



Queue: F → J → C → G

Visited: D, E



BFS Traversal

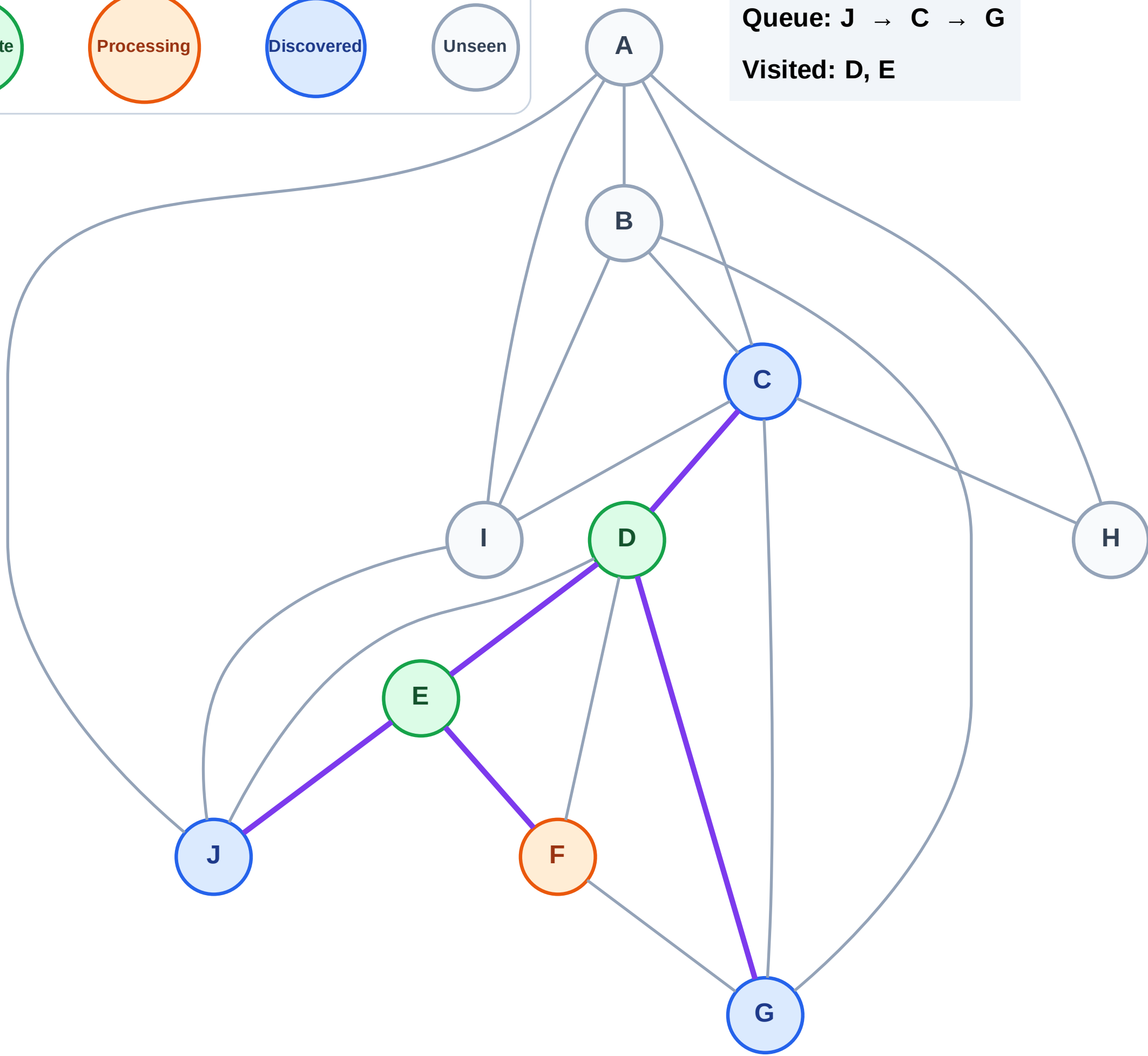
Step 6: Process node F

Legend



Queue: J → C → G

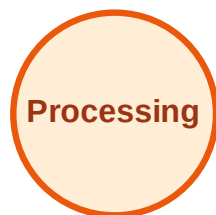
Visited: D, E



BFS Traversal

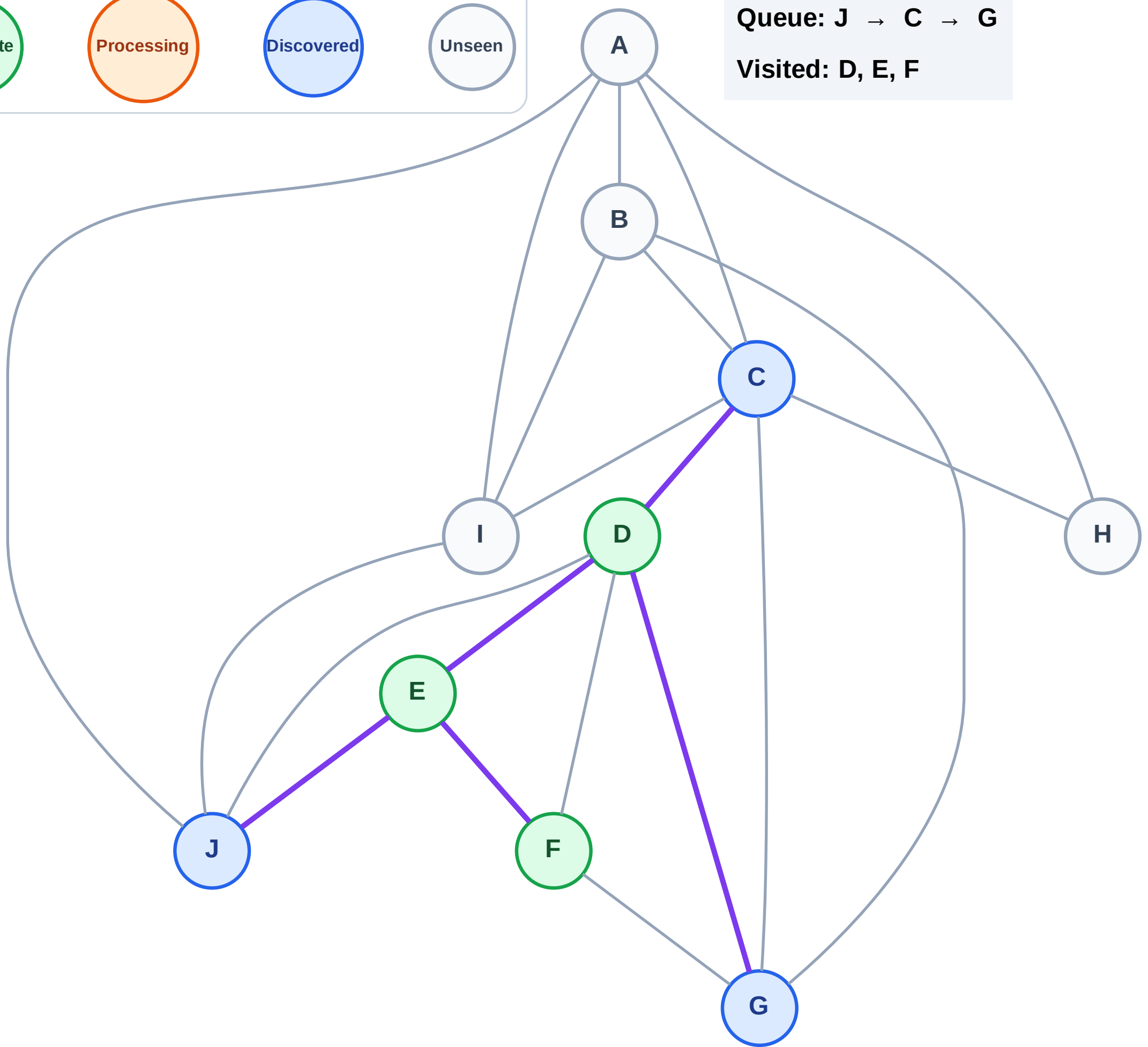
Step 7: No new neighbors from F

Legend



Queue: J → C → G

Visited: D, E, F



BFS Traversal

Step 8: Process node J

Legend

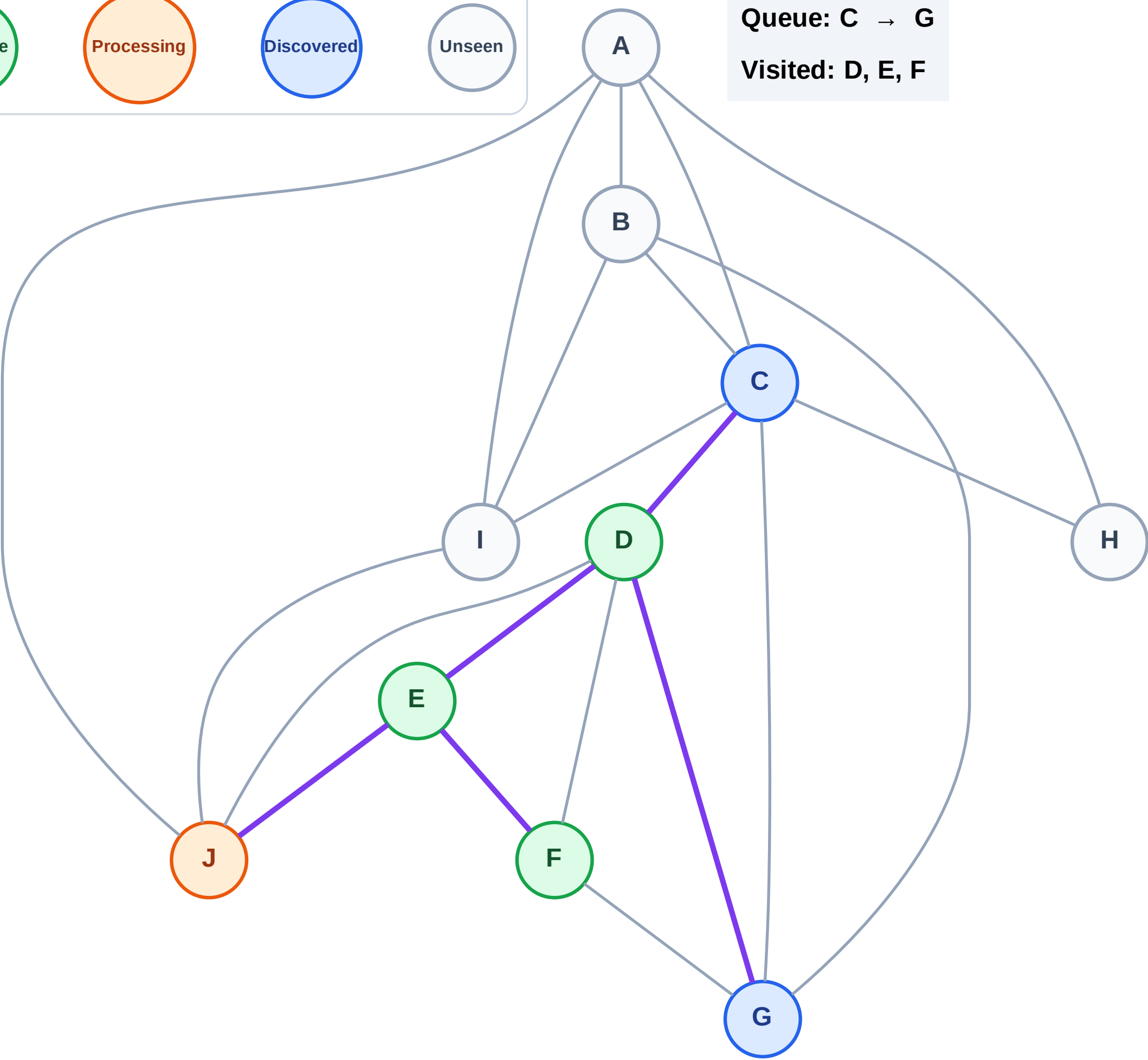
Complete

Processing

Discovered

Unseen

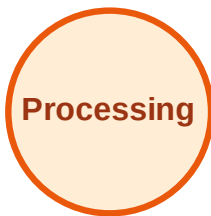
Queue: C → G
Visited: D, E, F



BFS Traversal

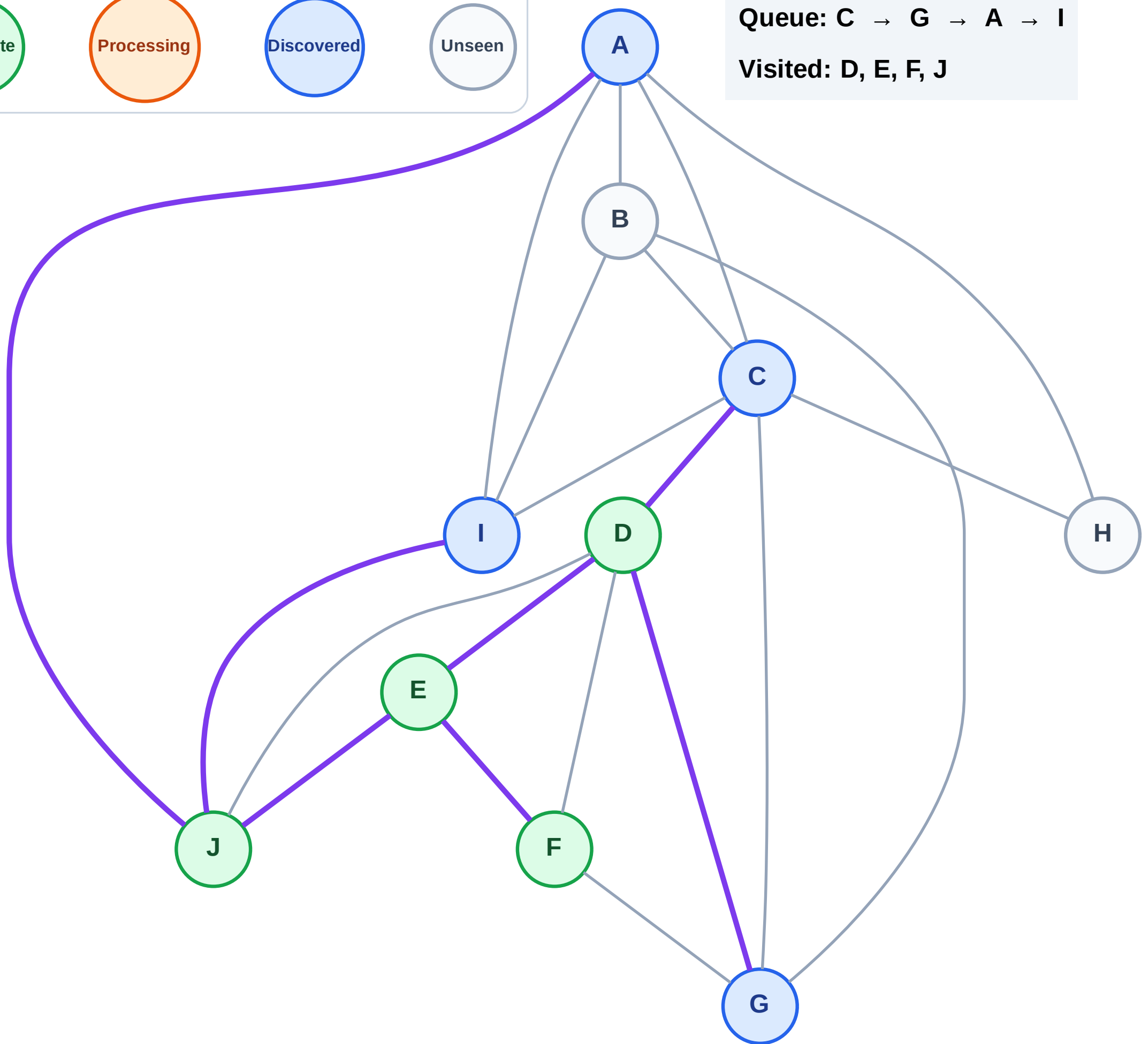
Step 9: Discover A, I from J

Legend



Queue: C → G → A → I

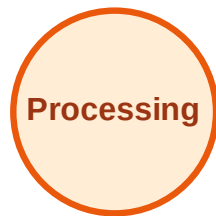
Visited: D, E, F, J



BFS Traversal

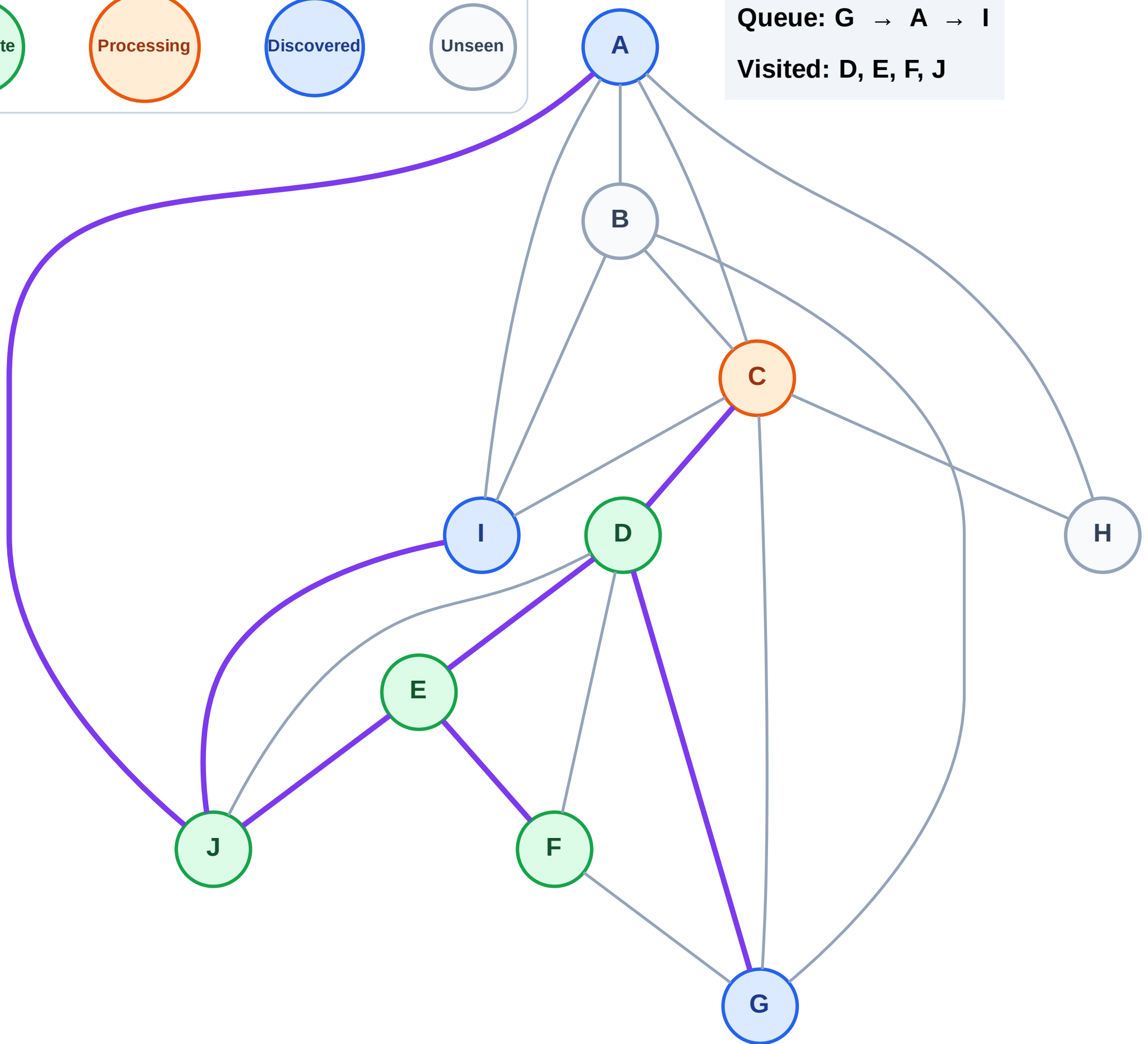
Step 10: Process node C

Legend



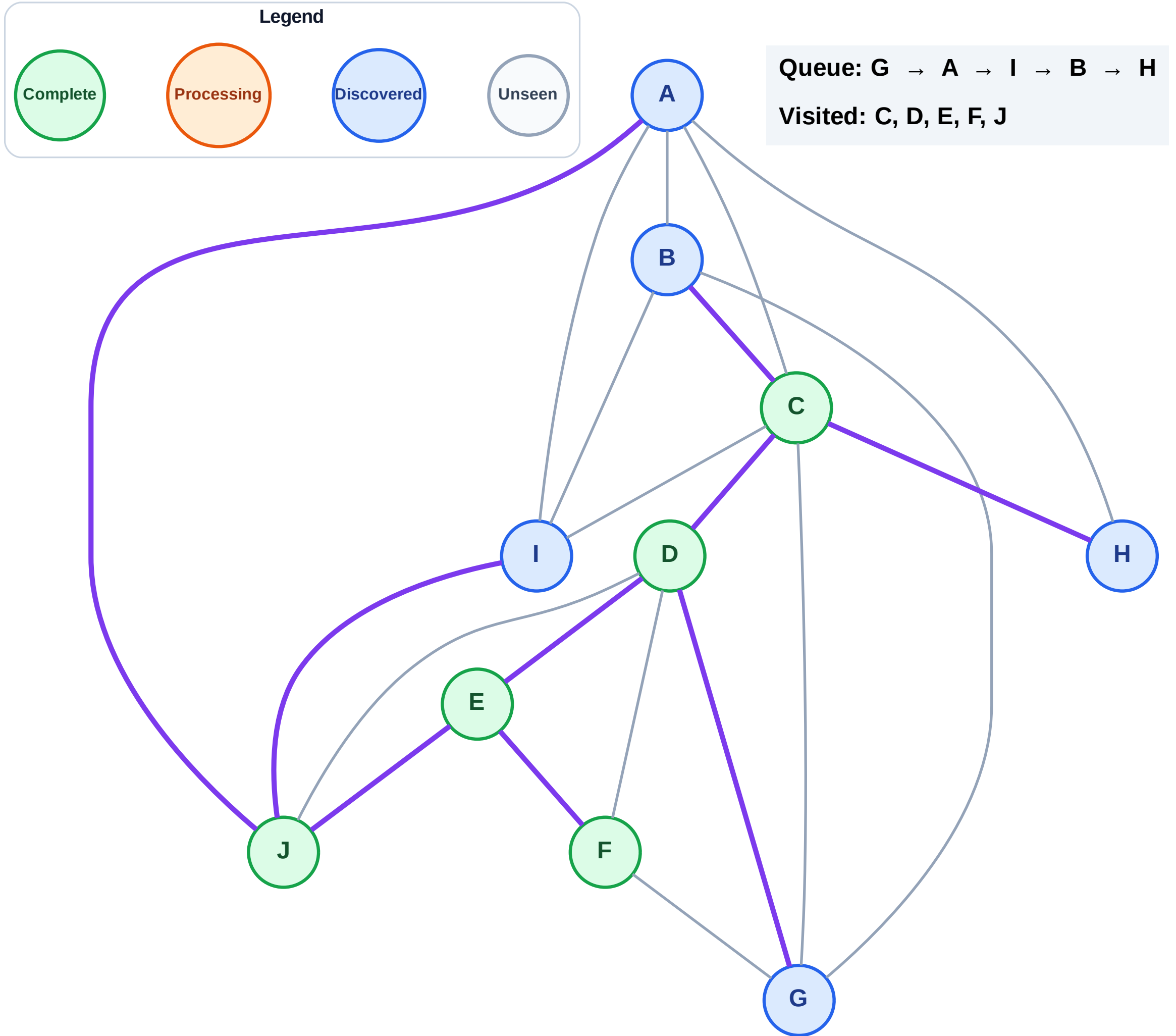
Queue: G → A → I

Visited: D, E, F, J



Step 11: Discover B, H from C

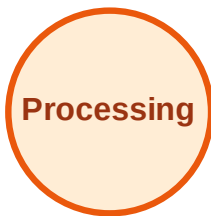
Step 11: Discover B, H from C



BFS Traversal

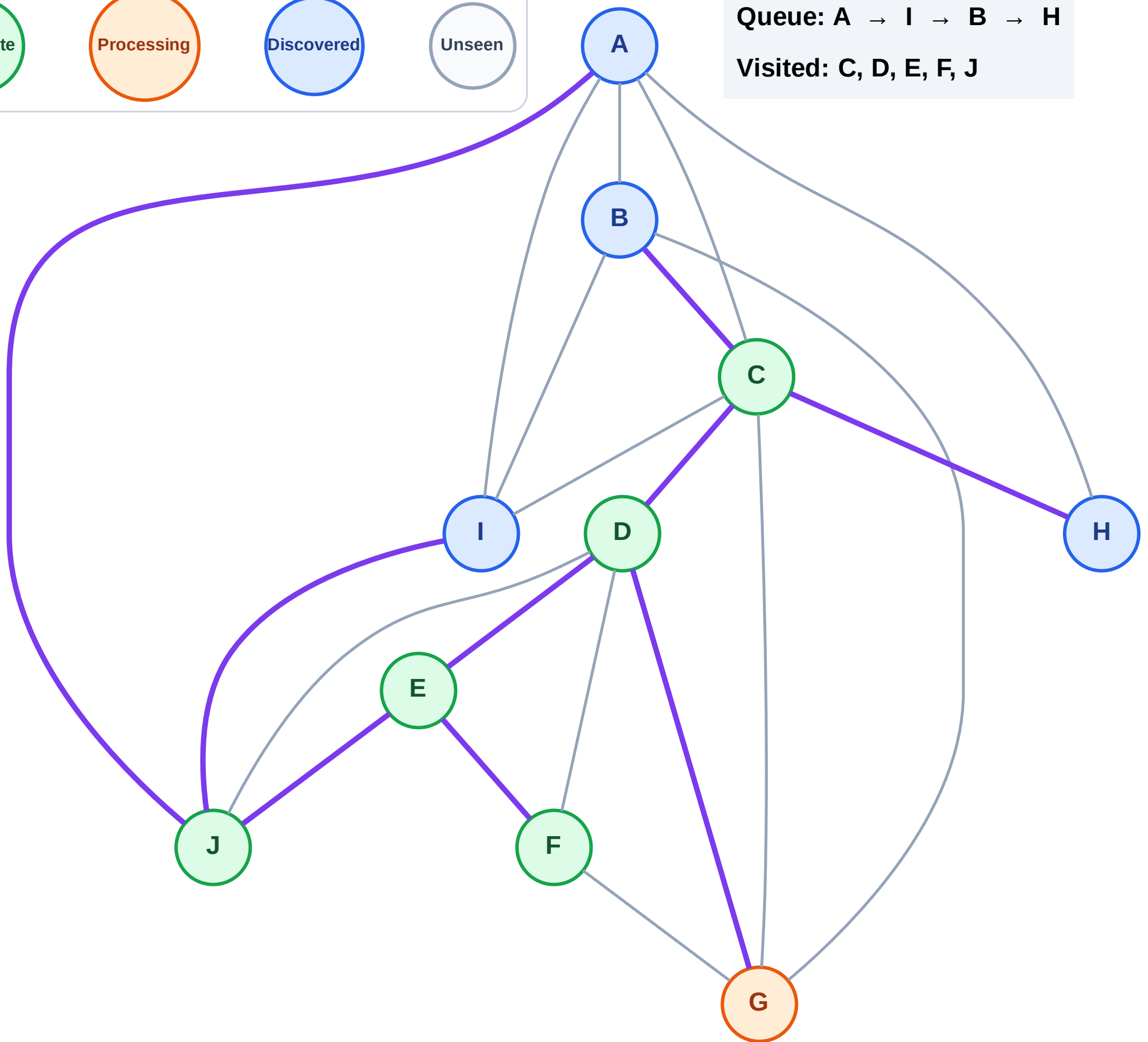
Step 12: Process node G

Legend



Queue: A → I → B → H

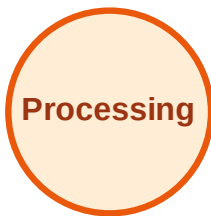
Visited: C, D, E, F, J



BFS Traversal

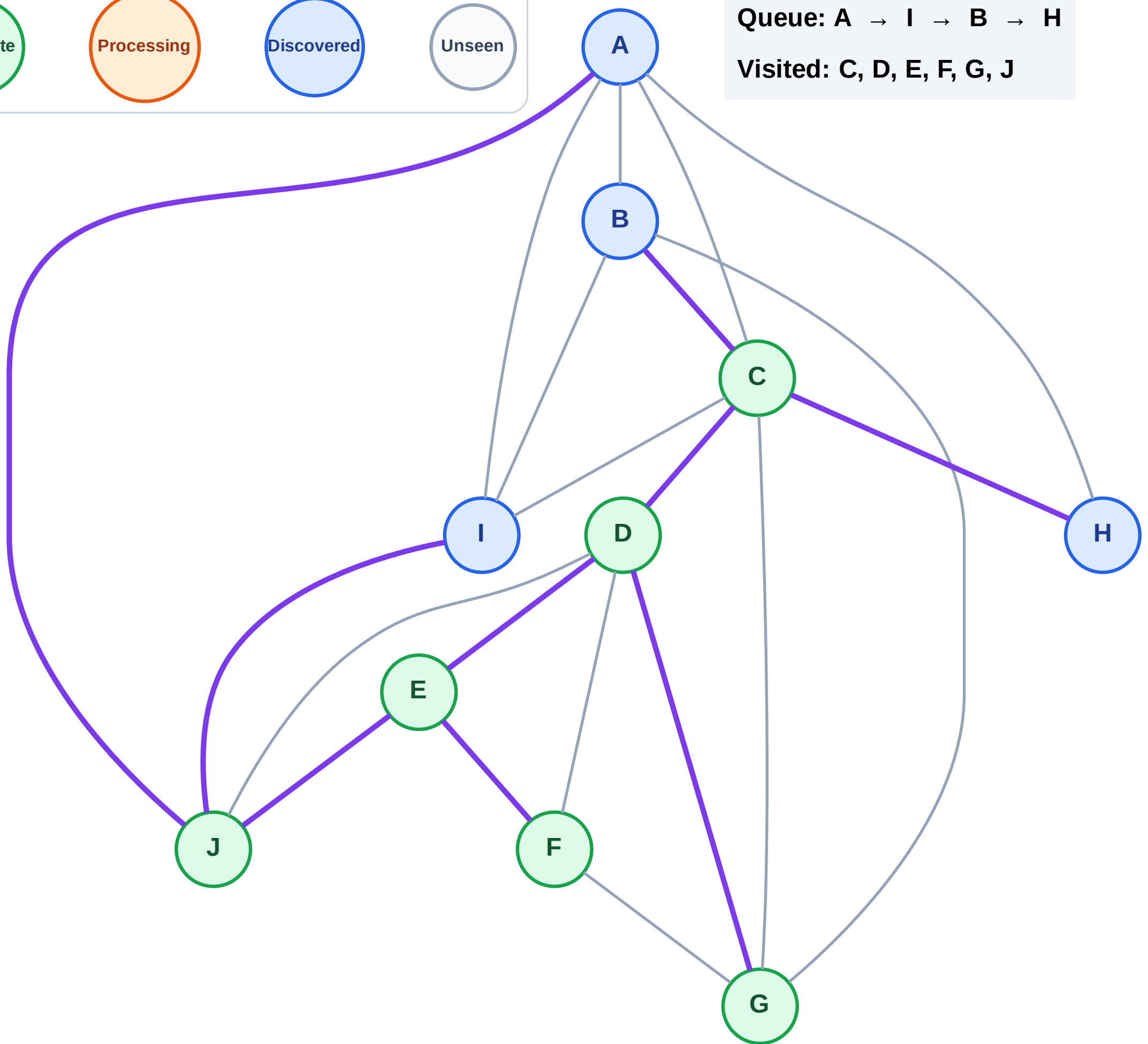
Step 13: No new neighbors from G

Legend



Queue: A → I → B → H

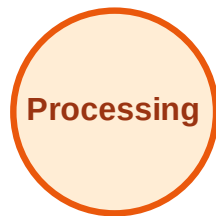
Visited: C, D, E, F, G, J



BFS Traversal

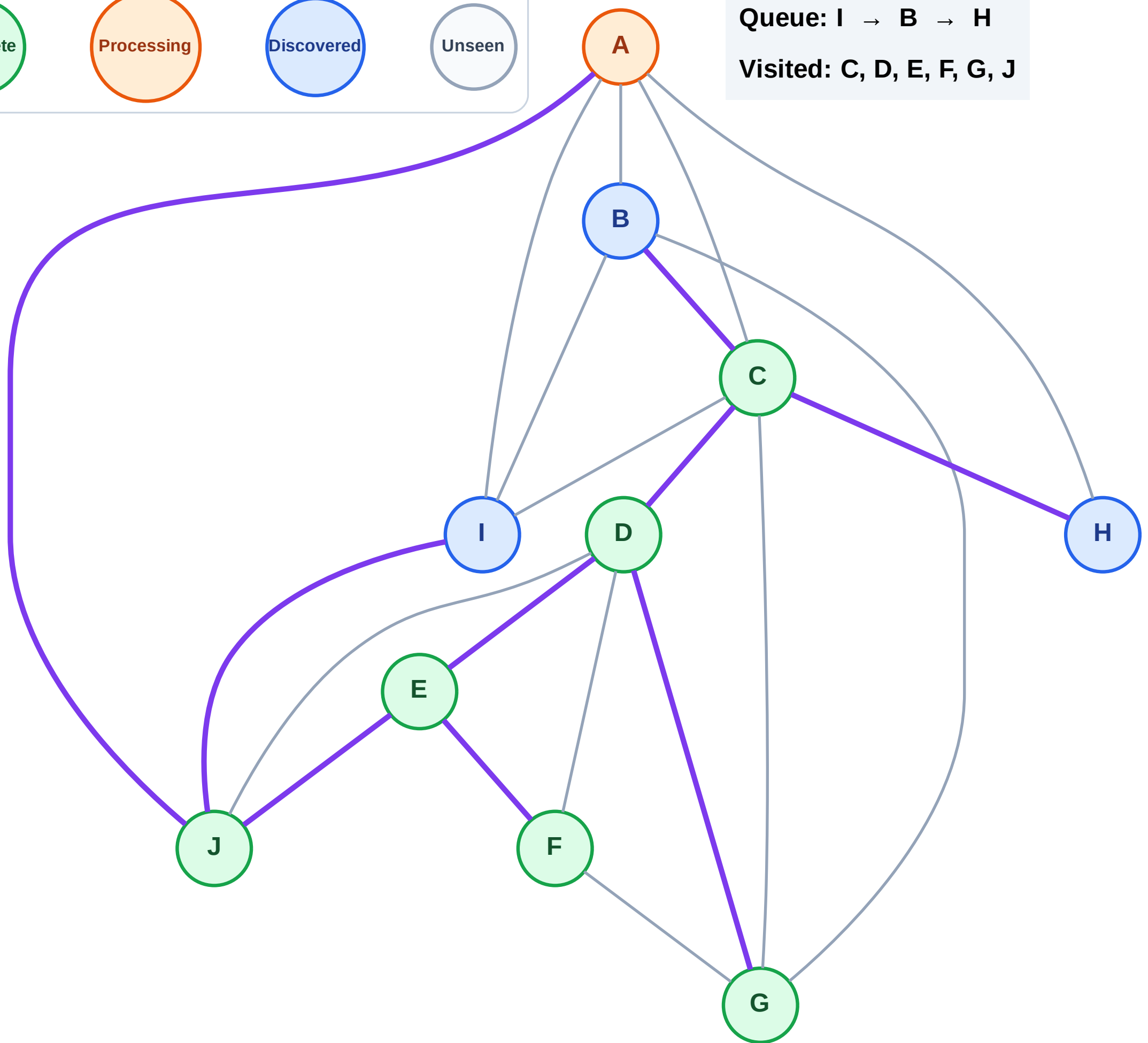
Step 14: Process node A

Legend



Queue: I → B → H

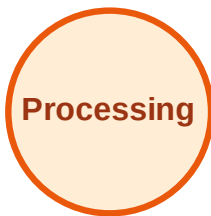
Visited: C, D, E, F, G, J



BFS Traversal

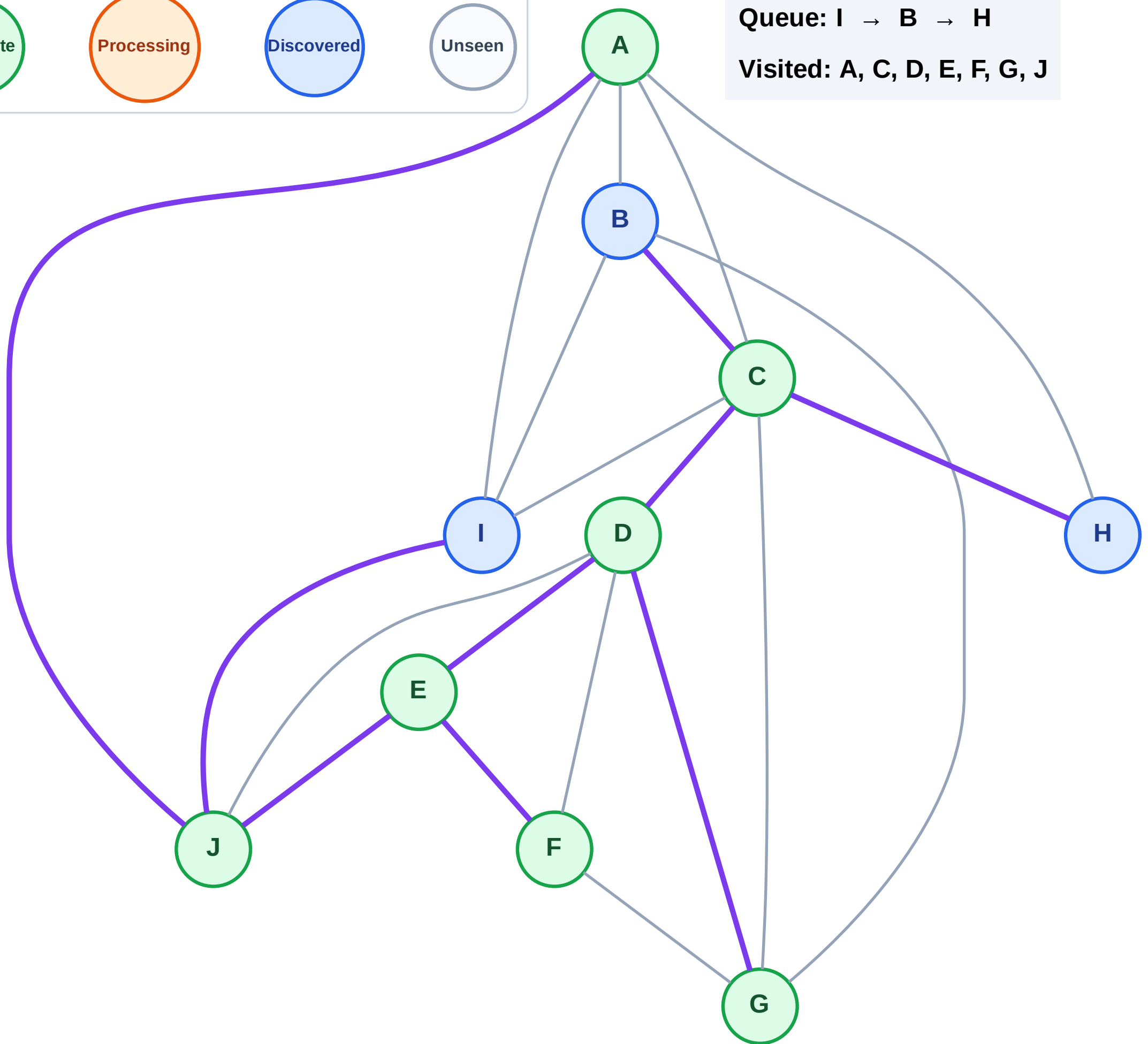
Step 15: No new neighbors from A

Legend



Queue: I → B → H

Visited: A, C, D, E, F, G, J



BFS Traversal

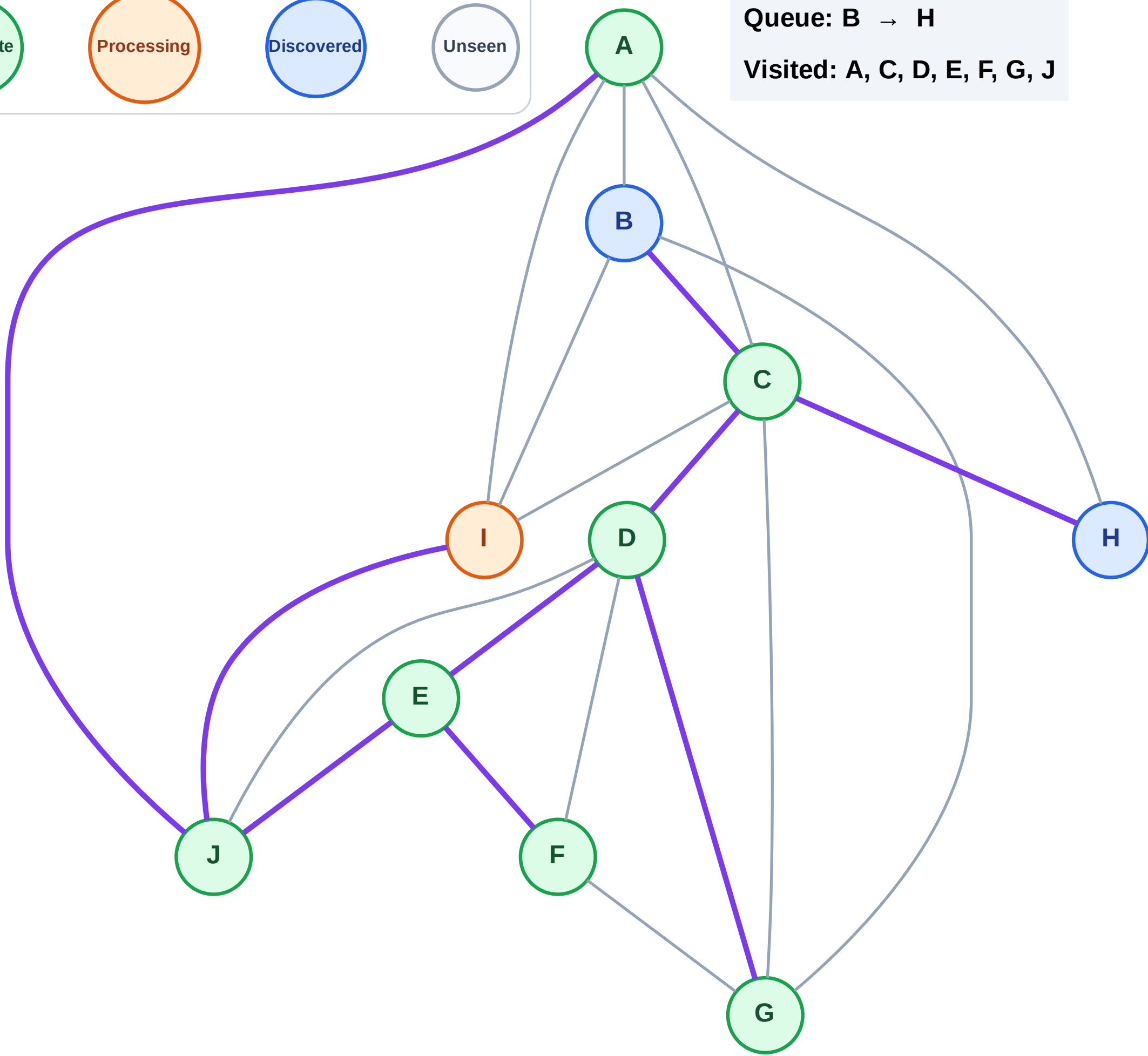
Step 16: Process node I

Legend



Queue: B → H

Visited: A, C, D, E, F, G, J



BFS Traversal

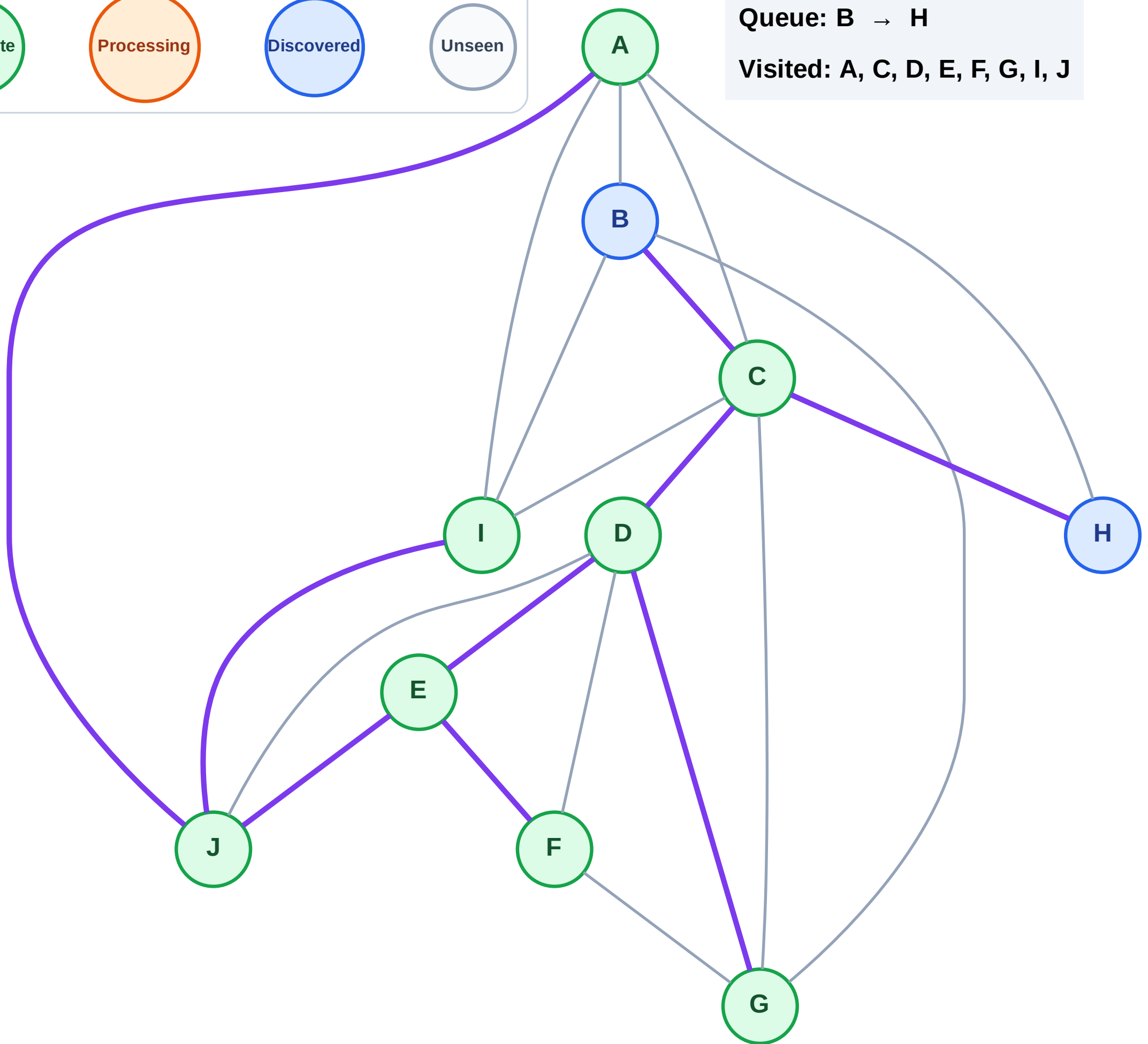
Step 17: No new neighbors from I

Legend



Queue: B → H

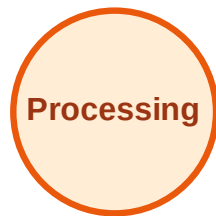
Visited: A, C, D, E, F, G, I, J



BFS Traversal

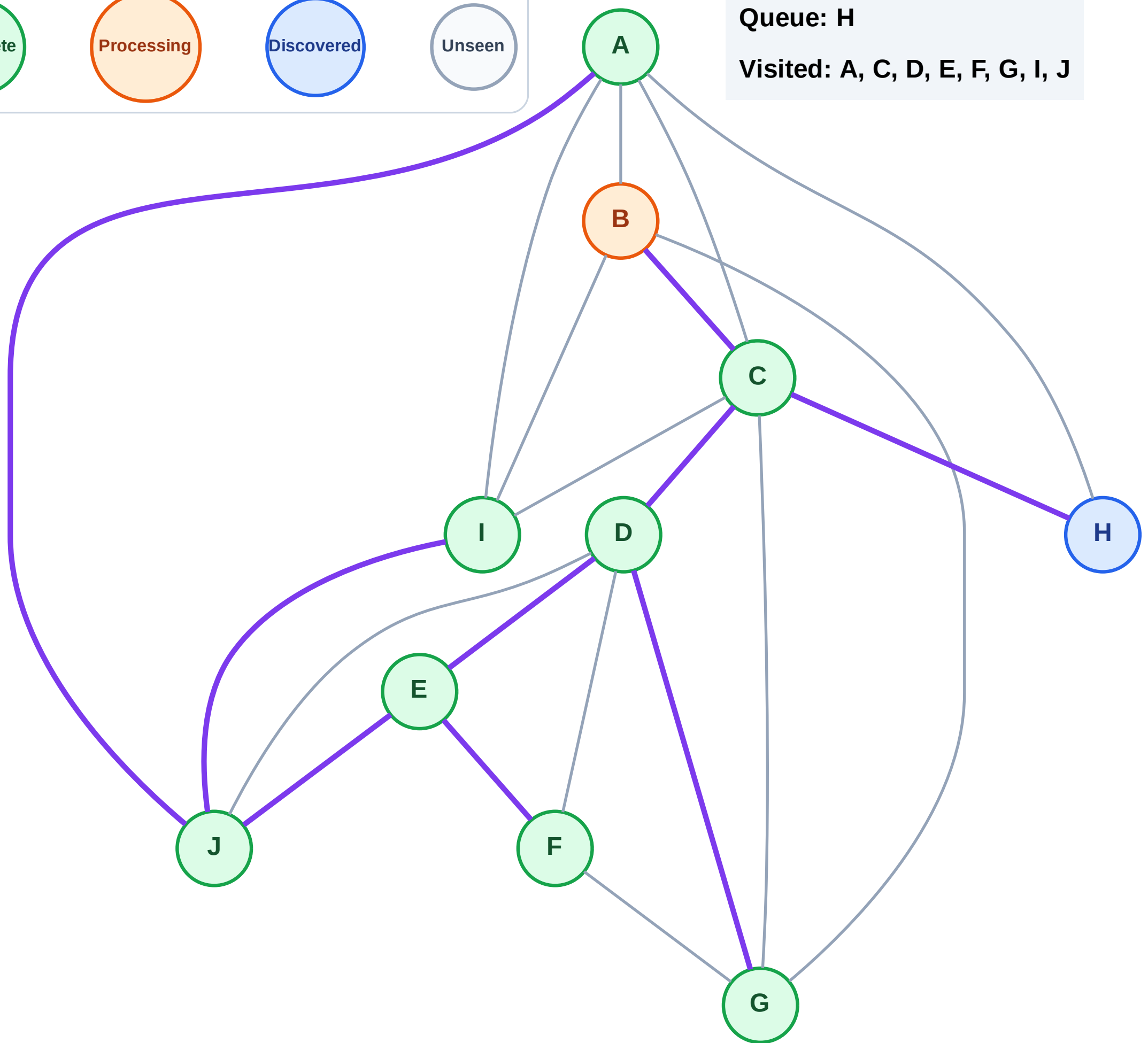
Step 18: Process node B

Legend



Queue: H

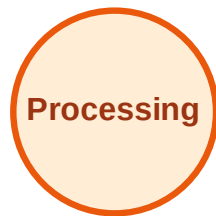
Visited: A, C, D, E, F, G, I, J



BFS Traversal

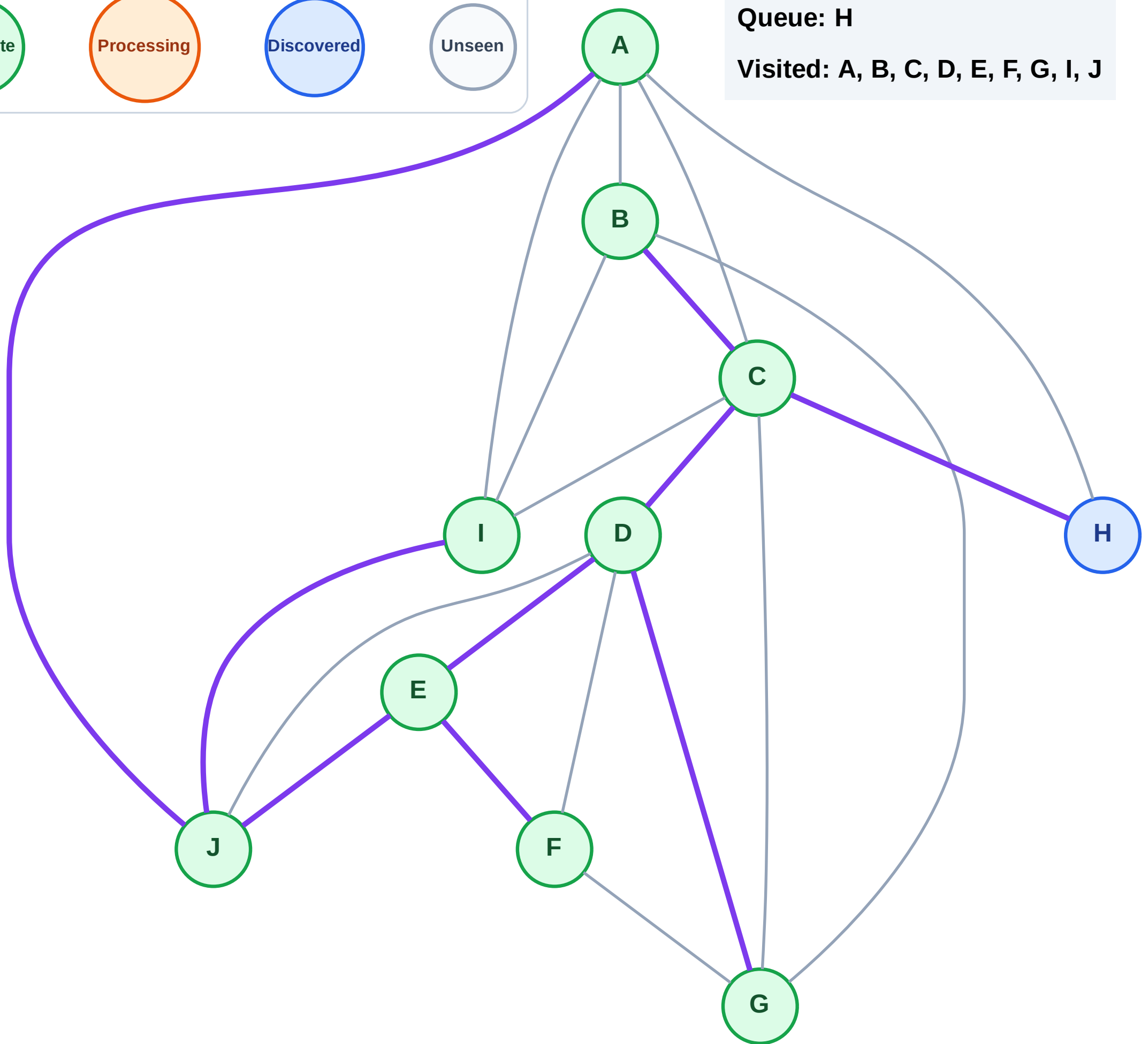
Step 19: No new neighbors from B

Legend



Queue: H

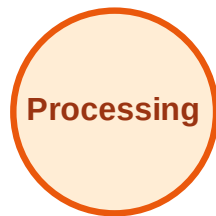
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

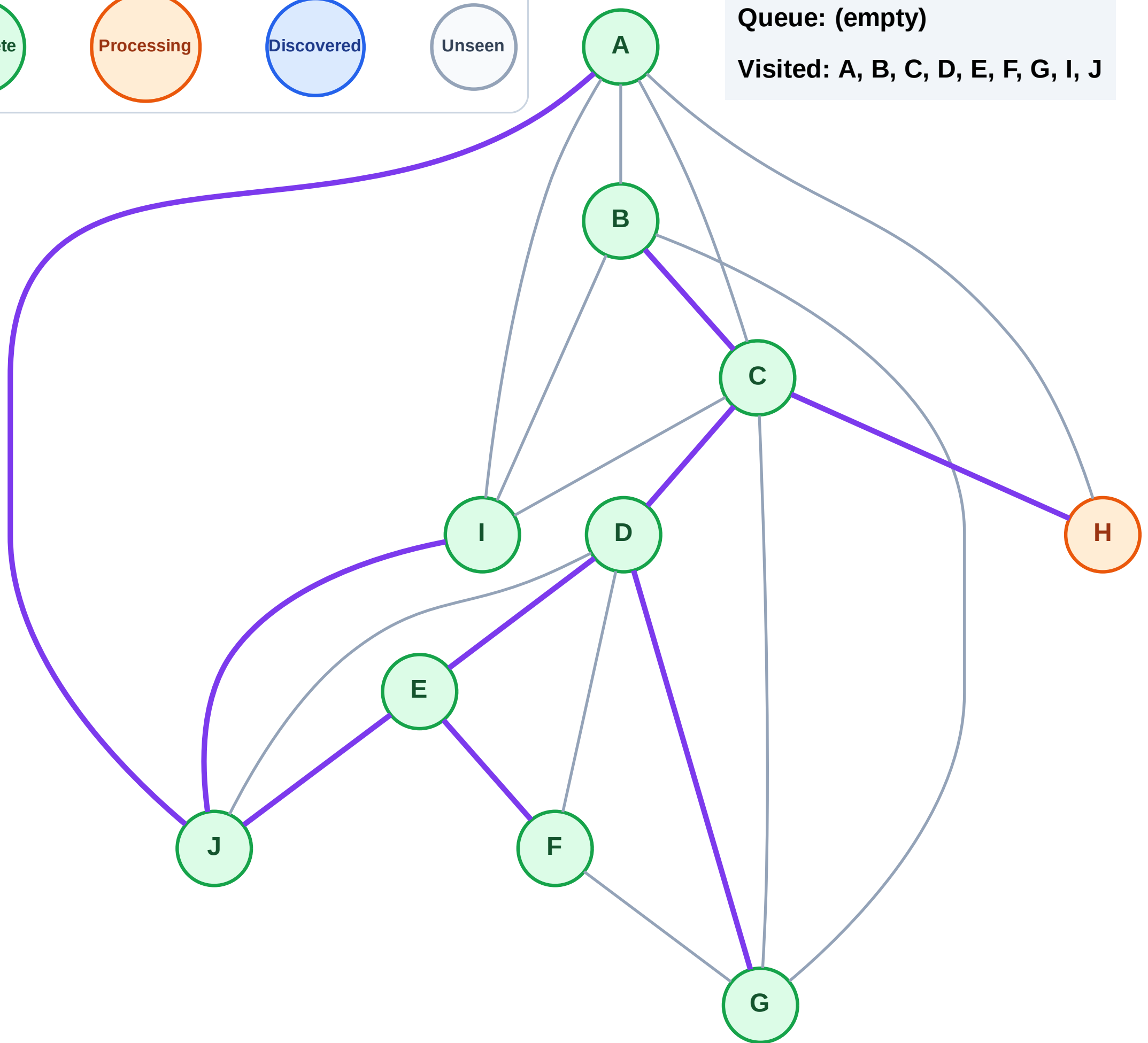
Step 20: Process node H

Legend



Queue: (empty)

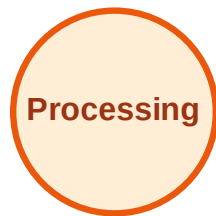
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

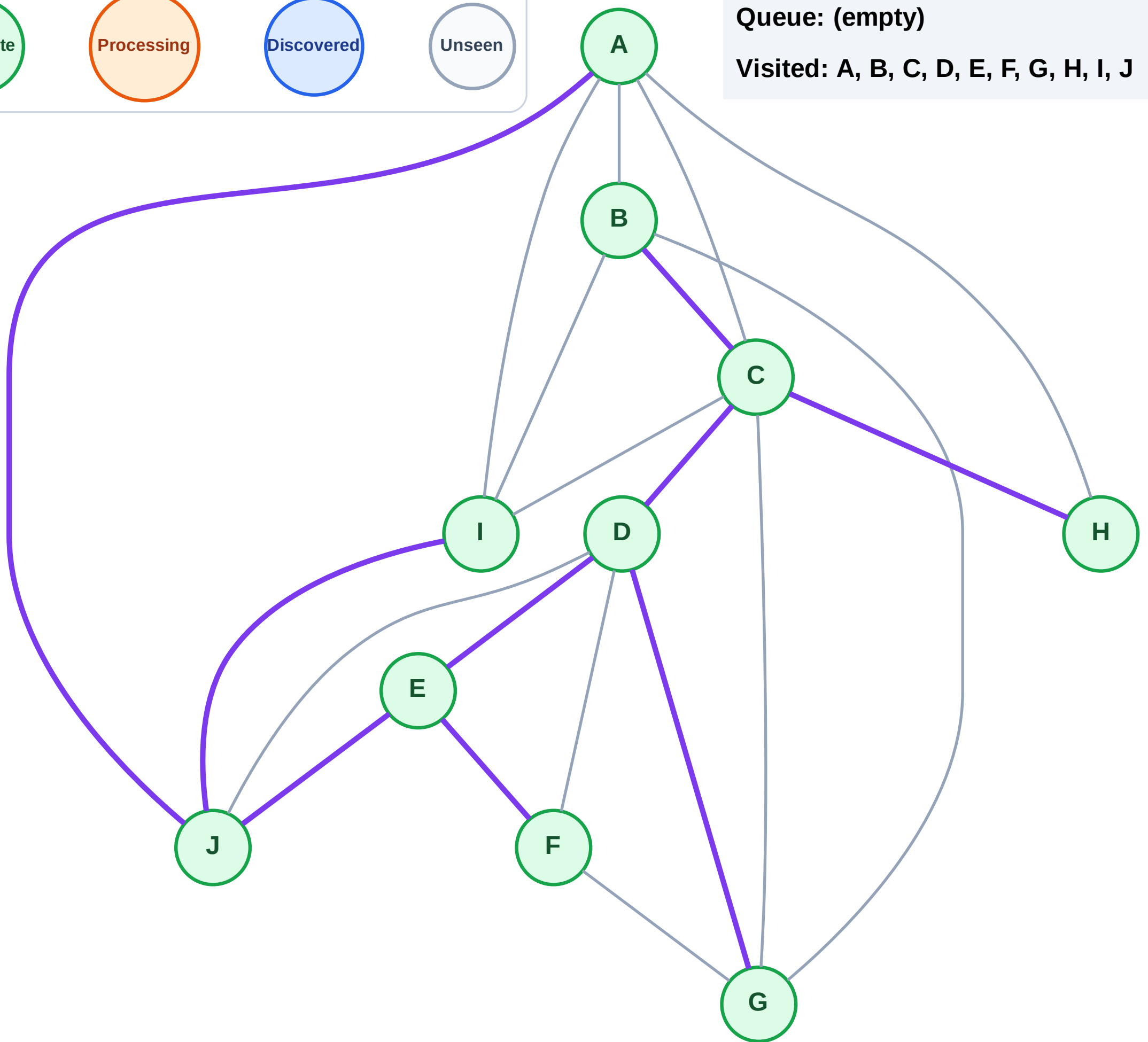
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

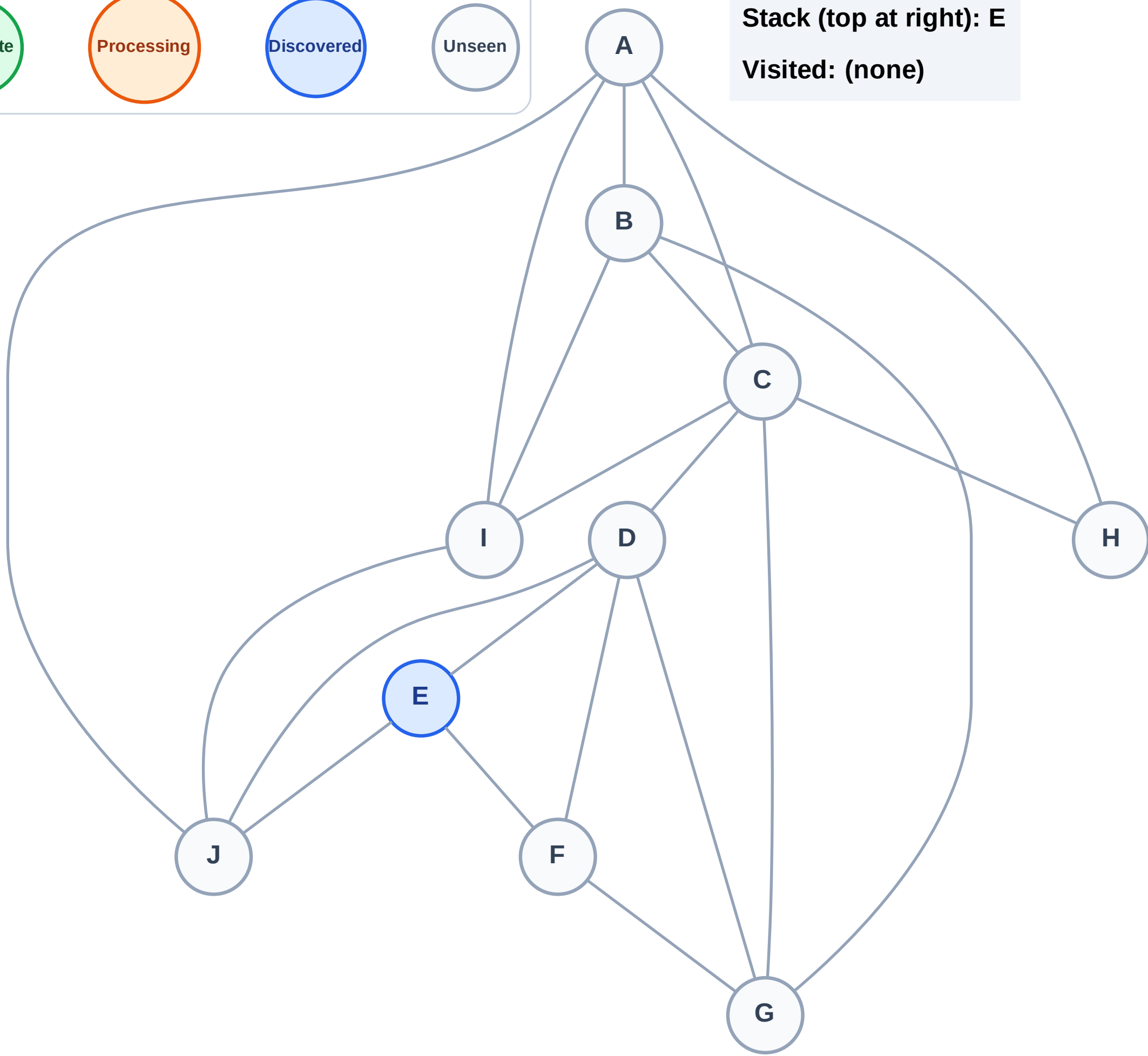
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

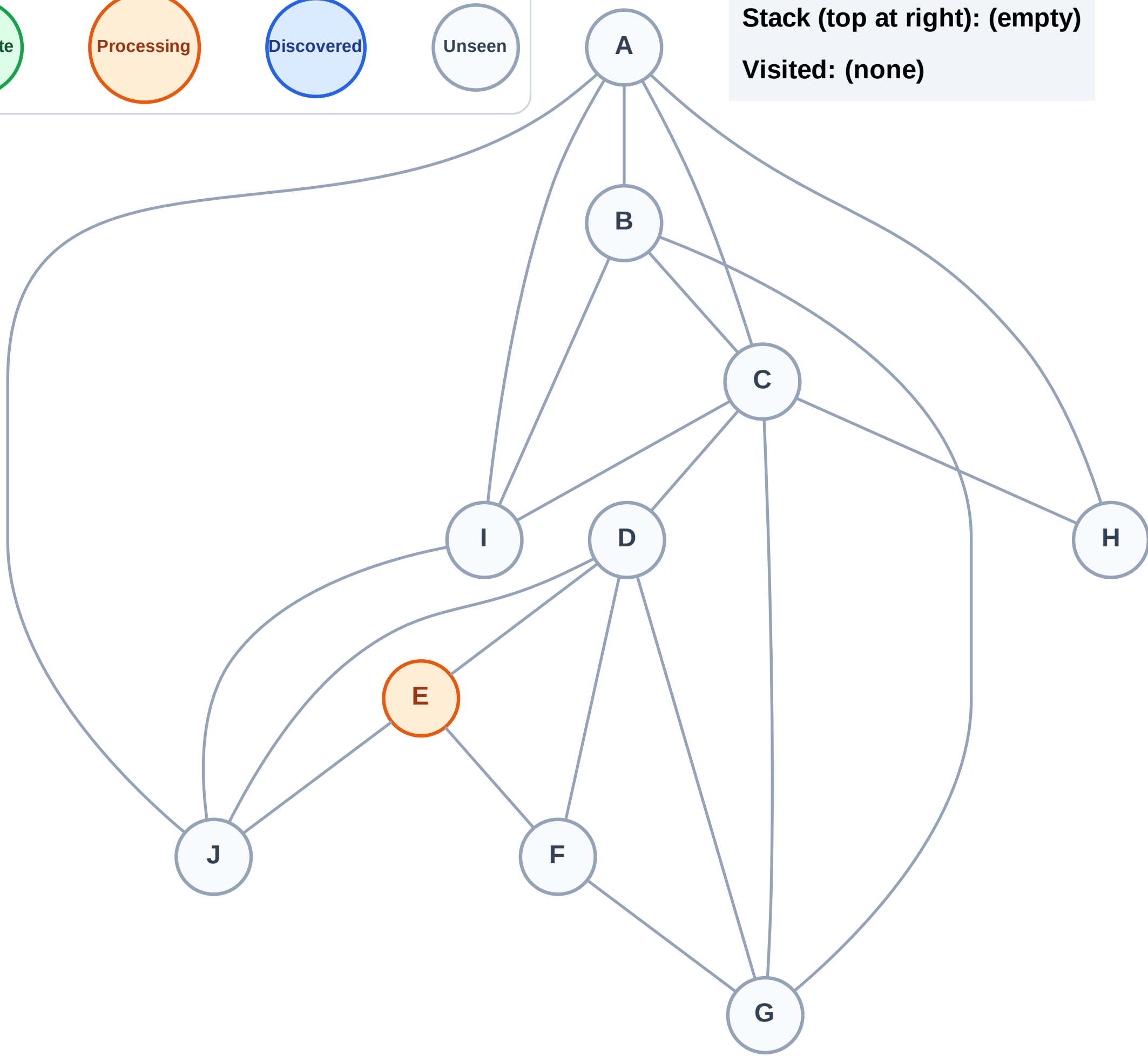
Step 2: Process node E

Legend



Stack (top at right): (empty)

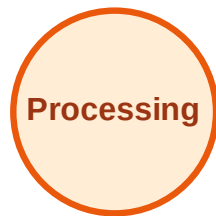
Visited: (none)



DFS Traversal

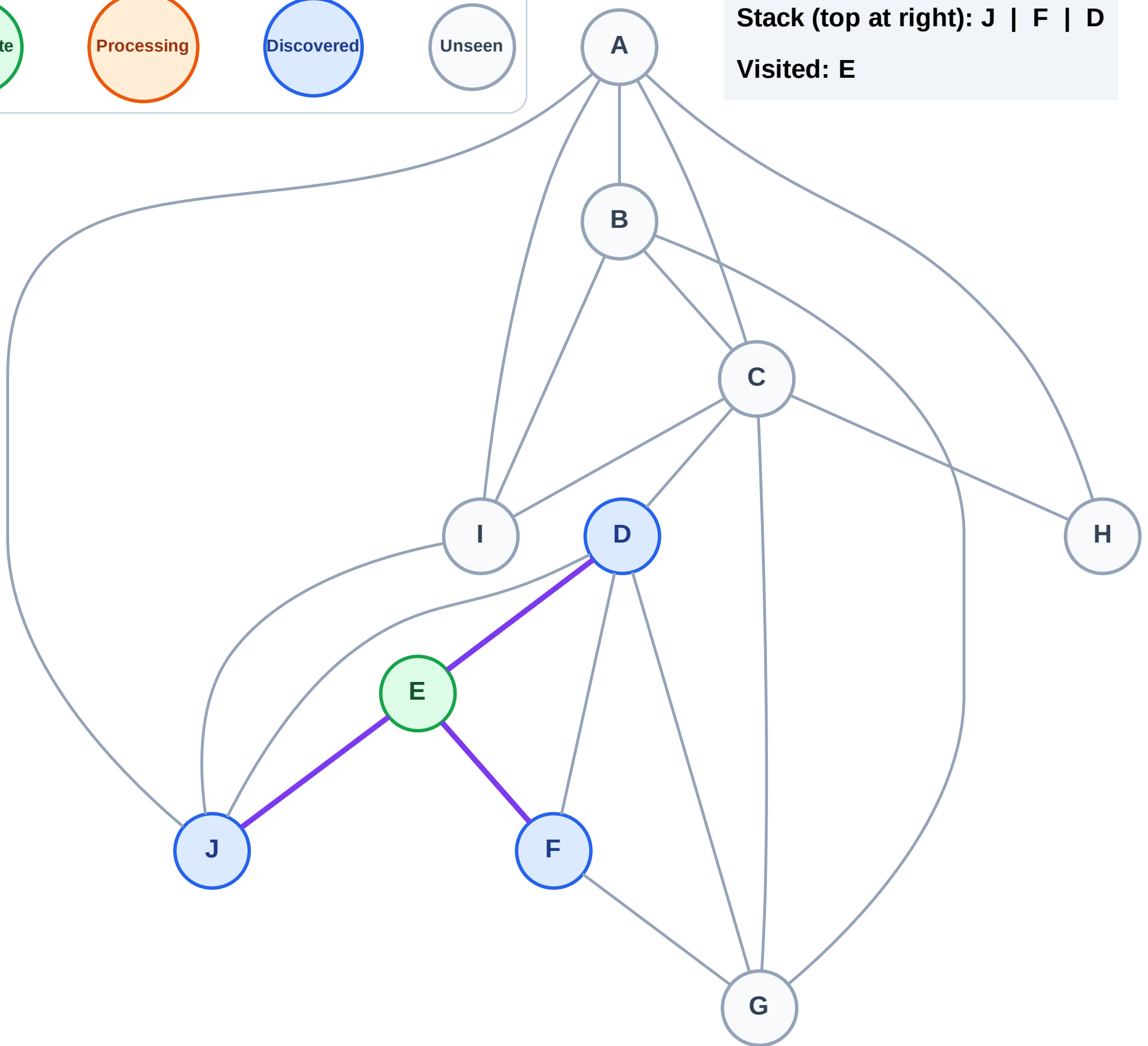
Step 3: Discover J, F, D from E

Legend



Stack (top at right): J | F | D

Visited: E



DFS Traversal

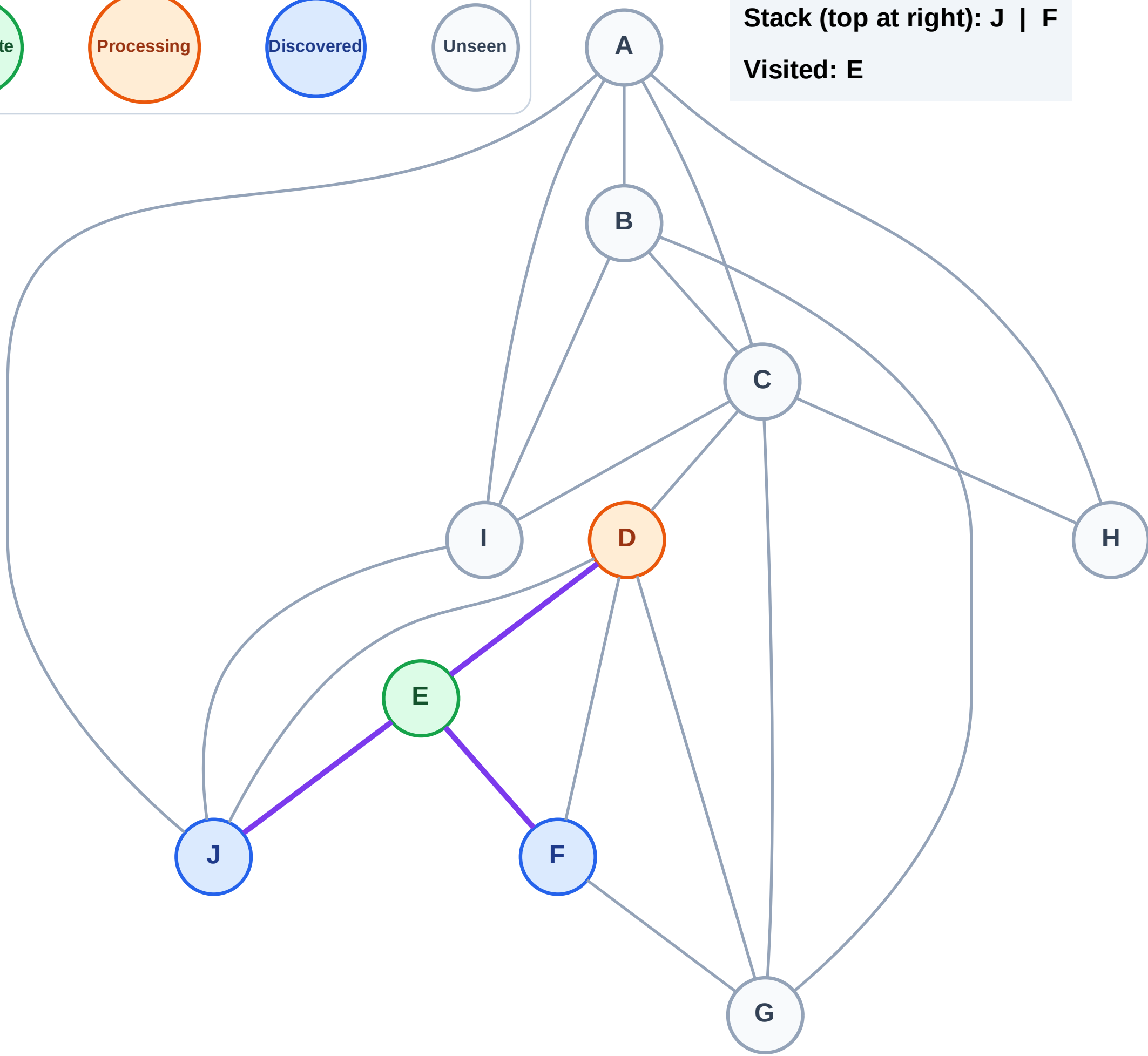
Step 4: Process node D

Legend



Stack (top at right): J | F

Visited: E



DFS Traversal

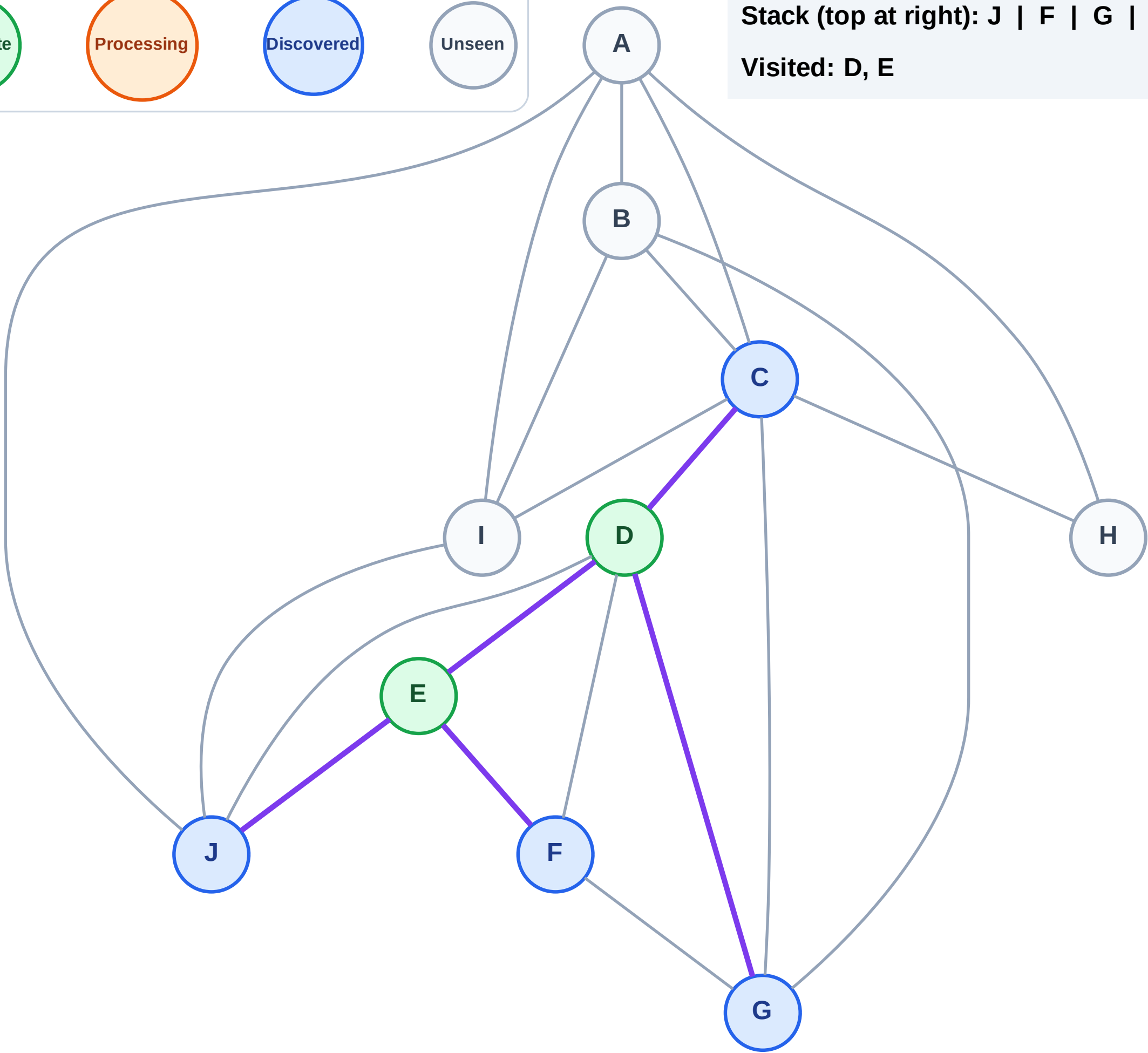
Step 5: Discover G, C from D

Legend



Stack (top at right): J | F | G | C

Visited: D, E



DFS Traversal

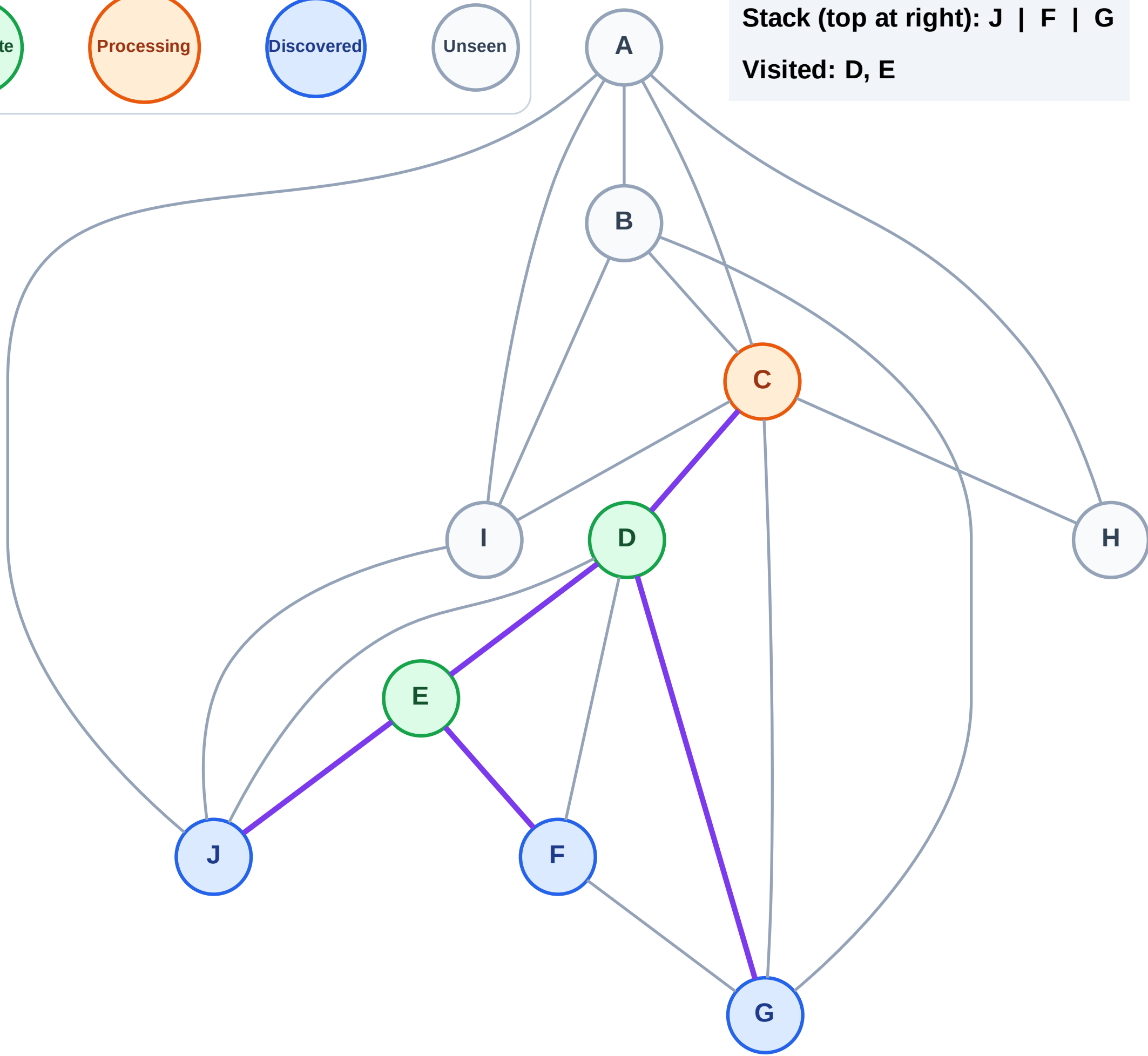
Step 6: Process node C

Legend



Stack (top at right): J | F | G

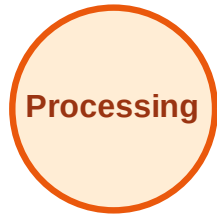
Visited: D, E



DFS Traversal

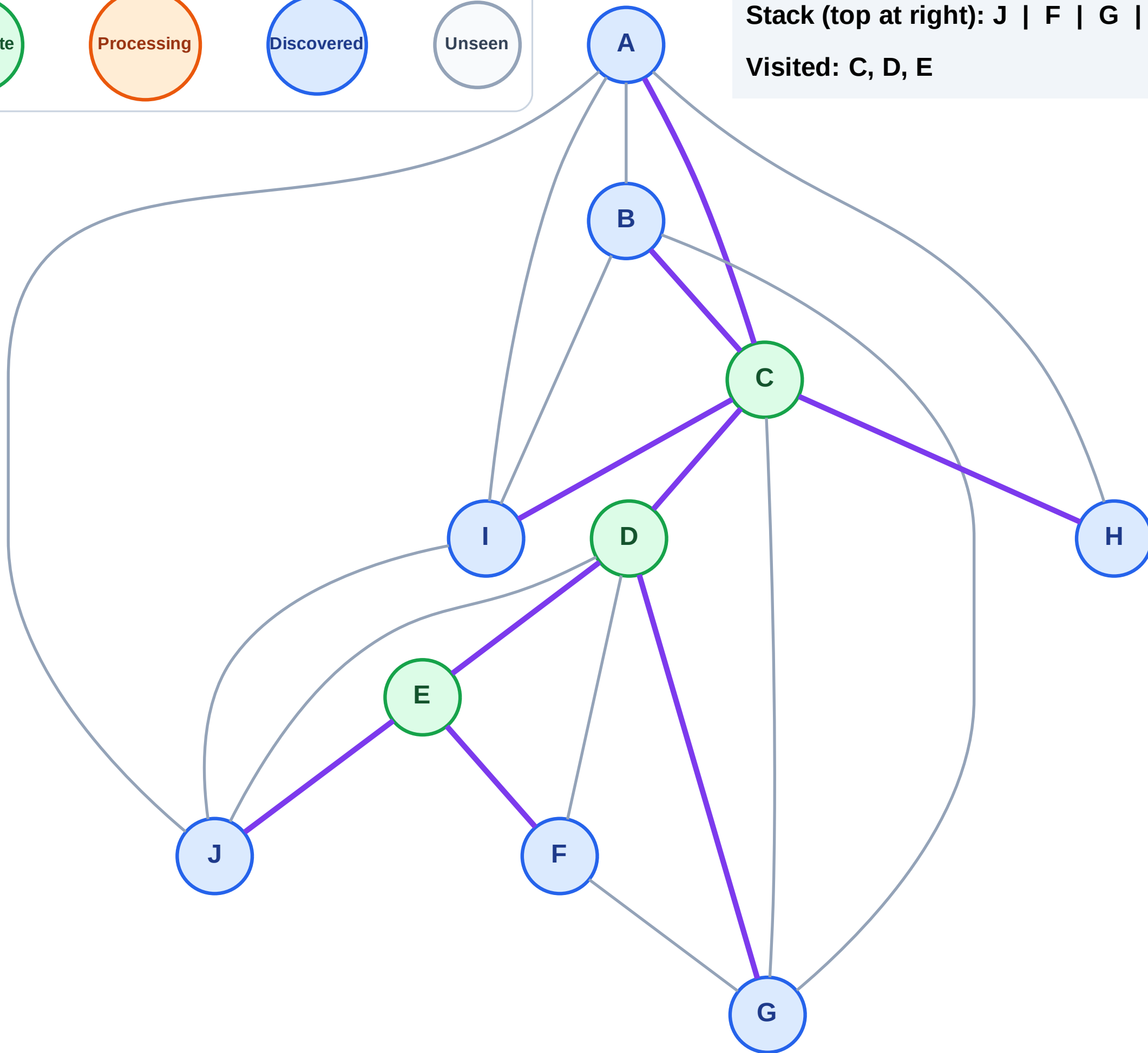
Step 7: Discover I, H, B, A from C

Legend



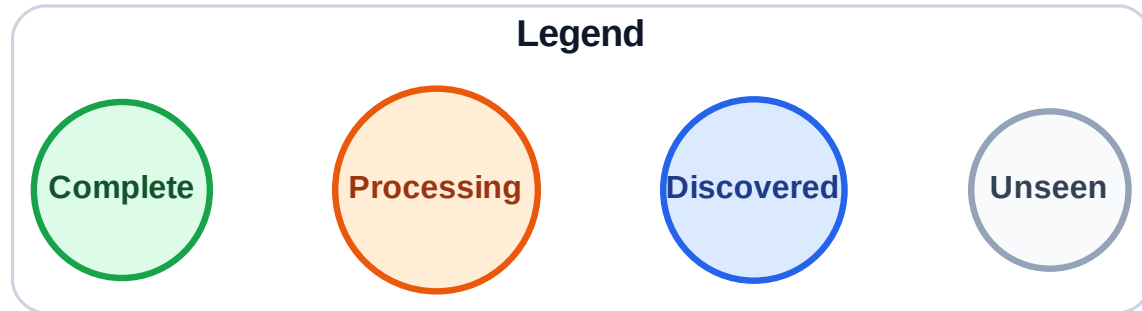
Stack (top at right): J | F | G | I | H | B | A

Visited: C, D, E



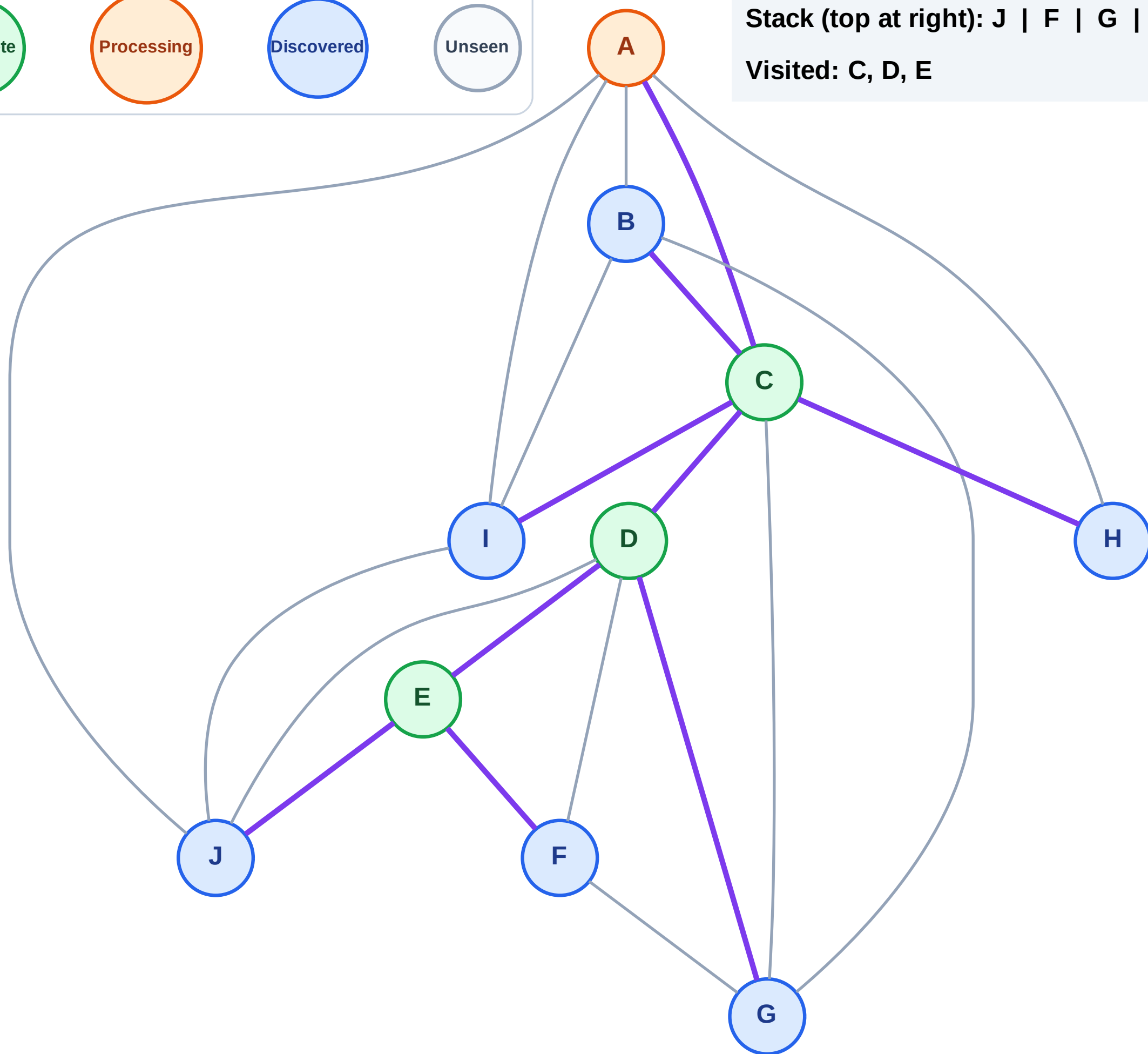
DFS Traversal

Step 8: Process node A



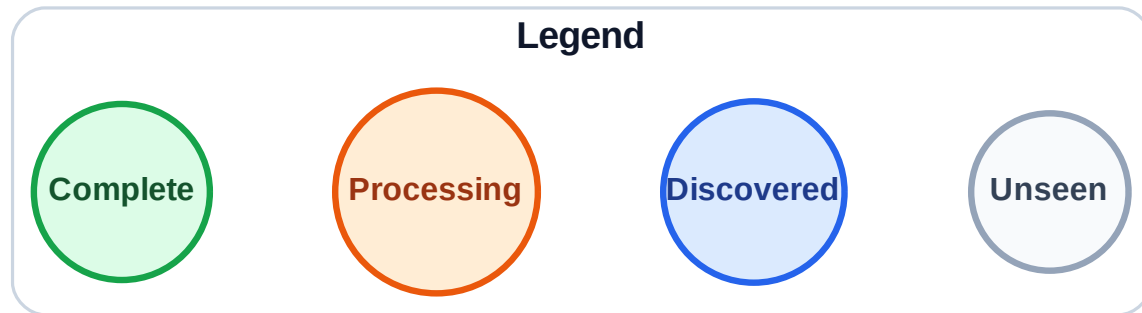
Stack (top at right): J | F | G | I | H | B

Visited: C, D, E

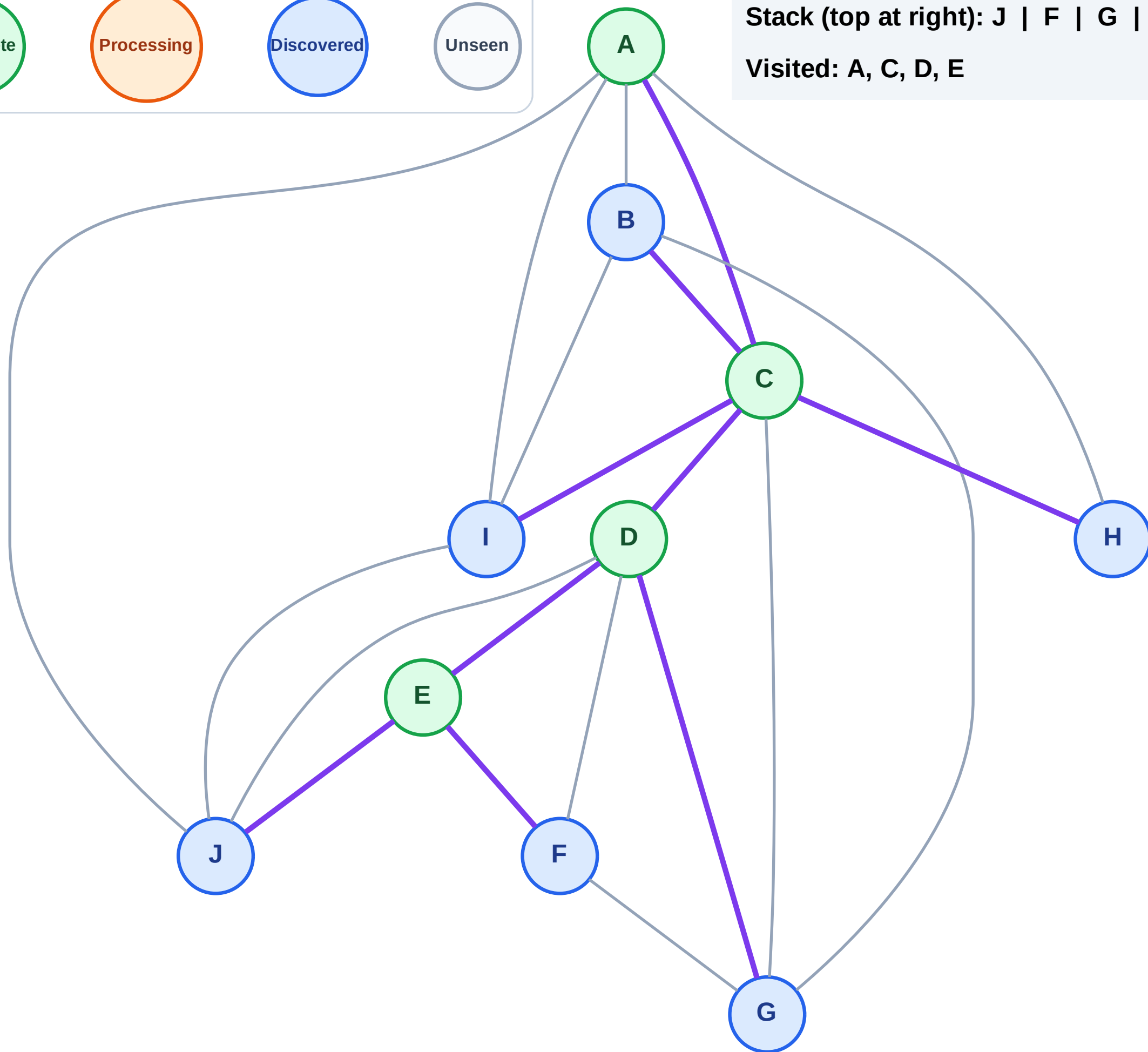


DFS Traversal

Step 9: No new neighbors from A



Stack (top at right): J | F | G | I | H | B
Visited: A, C, D, E



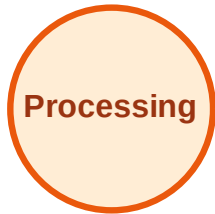
DFS Traversal

Step 10: Process node B

Legend



Complete



Processing



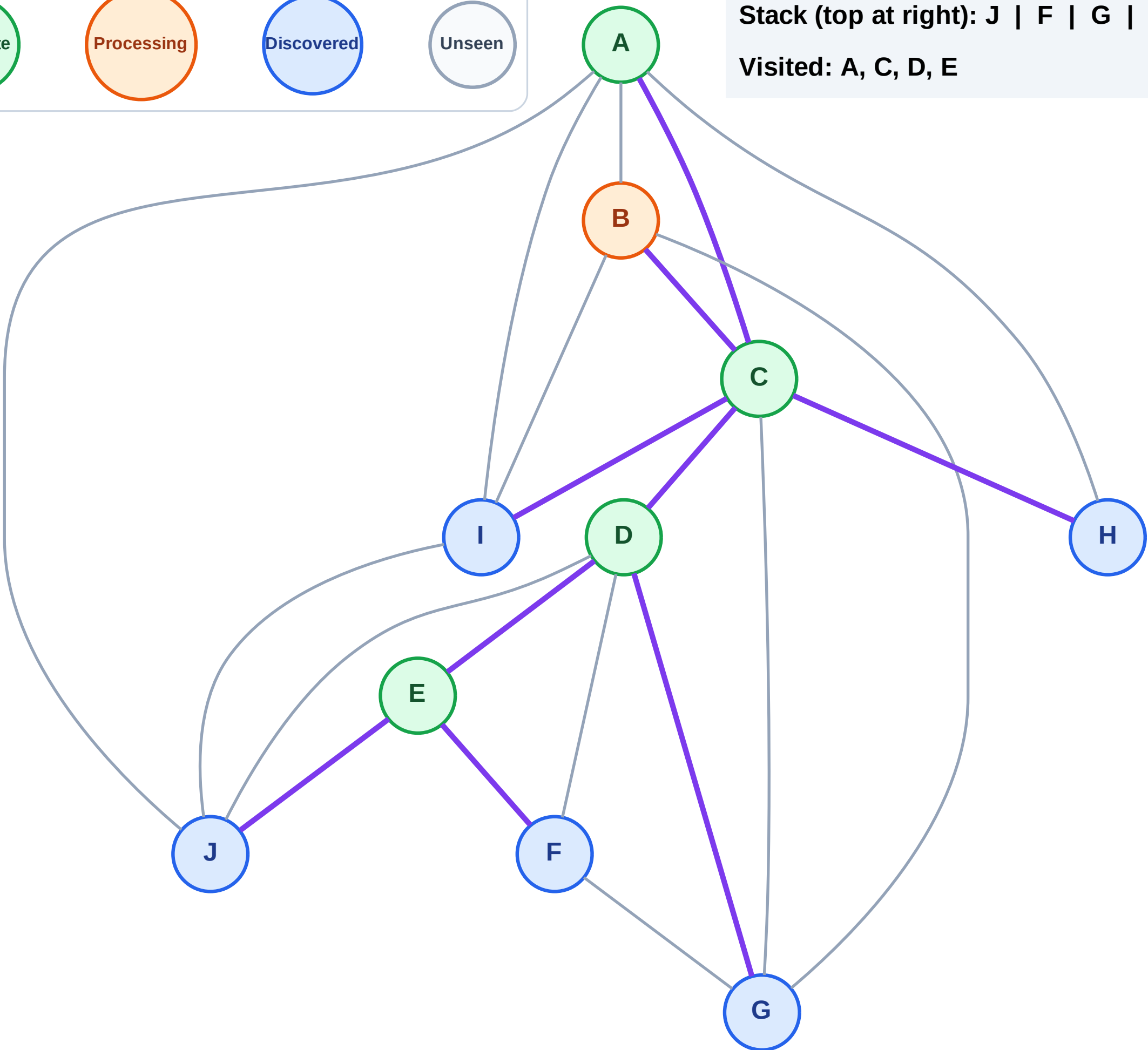
Discovered



Unseen

Stack (top at right): J | F | G | I | H

Visited: A, C, D, E

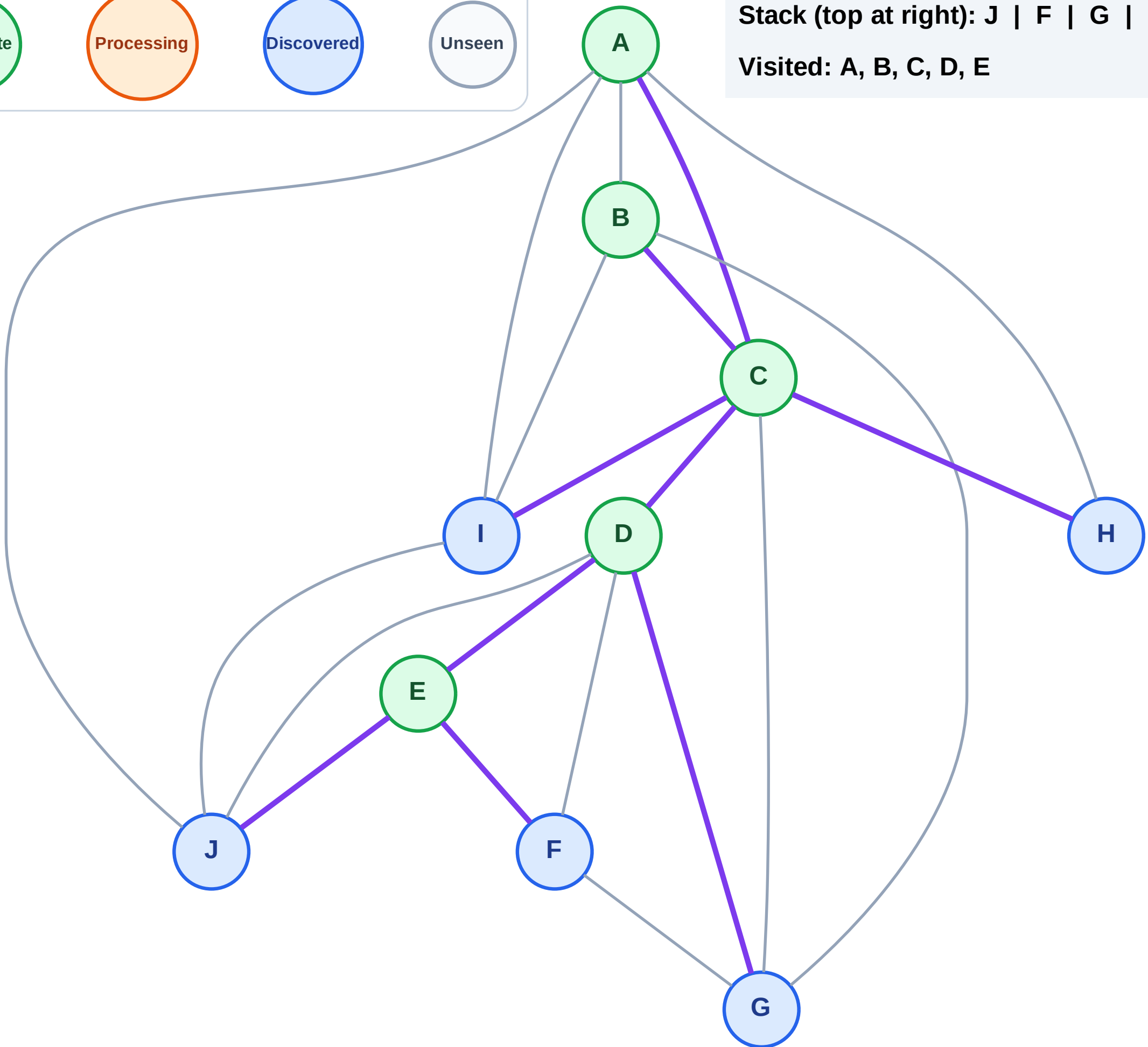
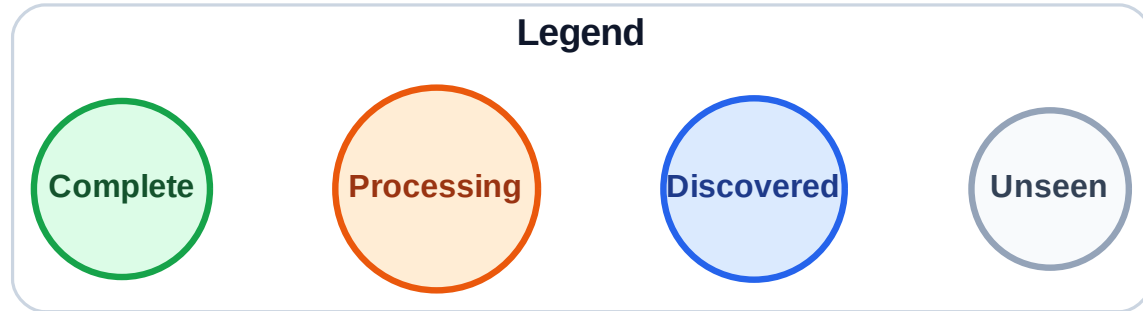


DFS Traversal

Step 11: No new neighbors from B

Stack (top at right): J | F | G | I | H

Visited: A, B, C, D, E



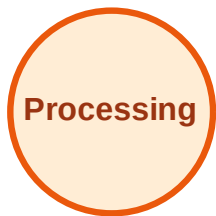
DFS Traversal

Step 12: Process node H

Legend



Complete



Processing



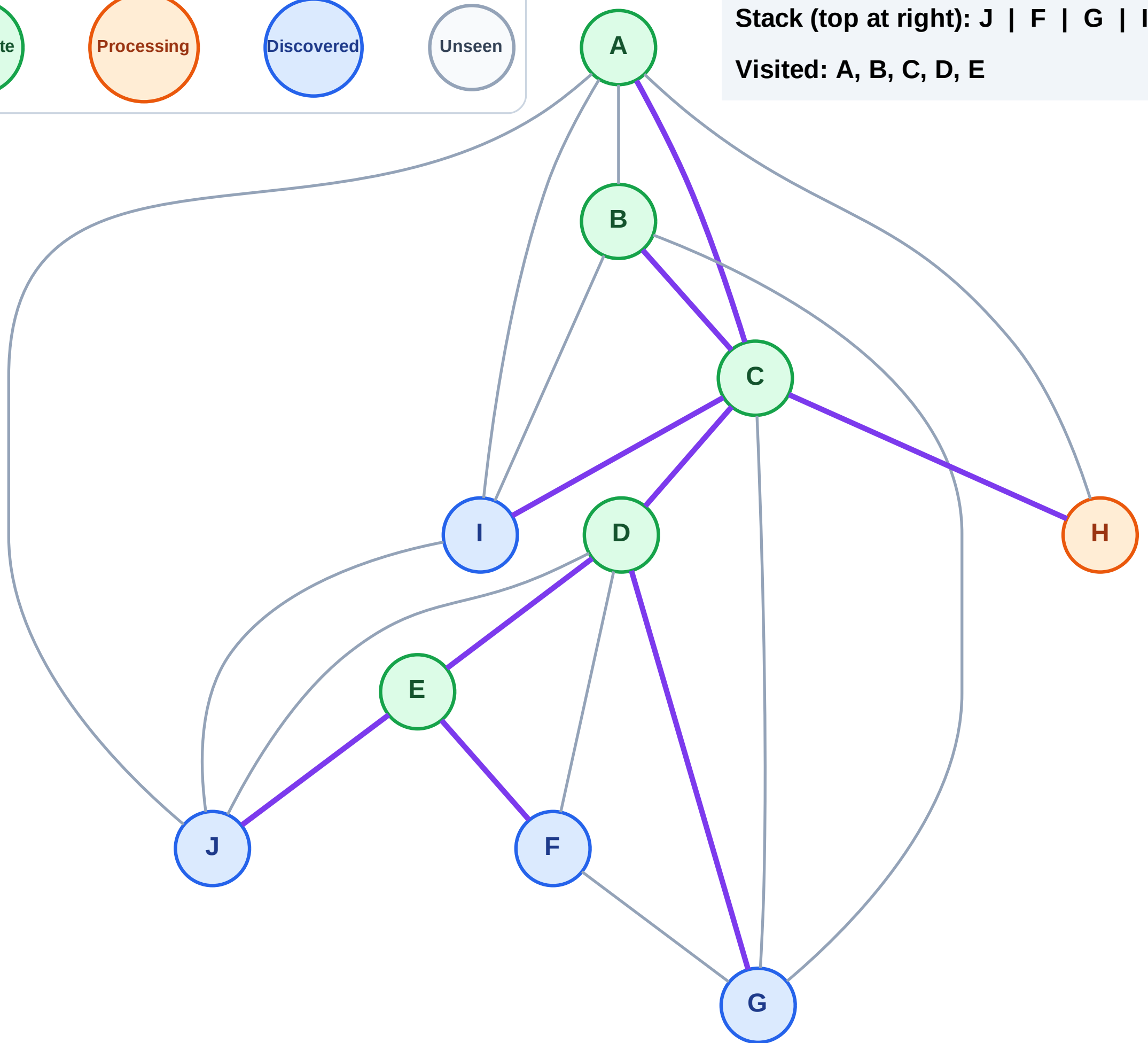
Discovered



Unseen

Stack (top at right): J | F | G | I

Visited: A, B, C, D, E



DFS Traversal

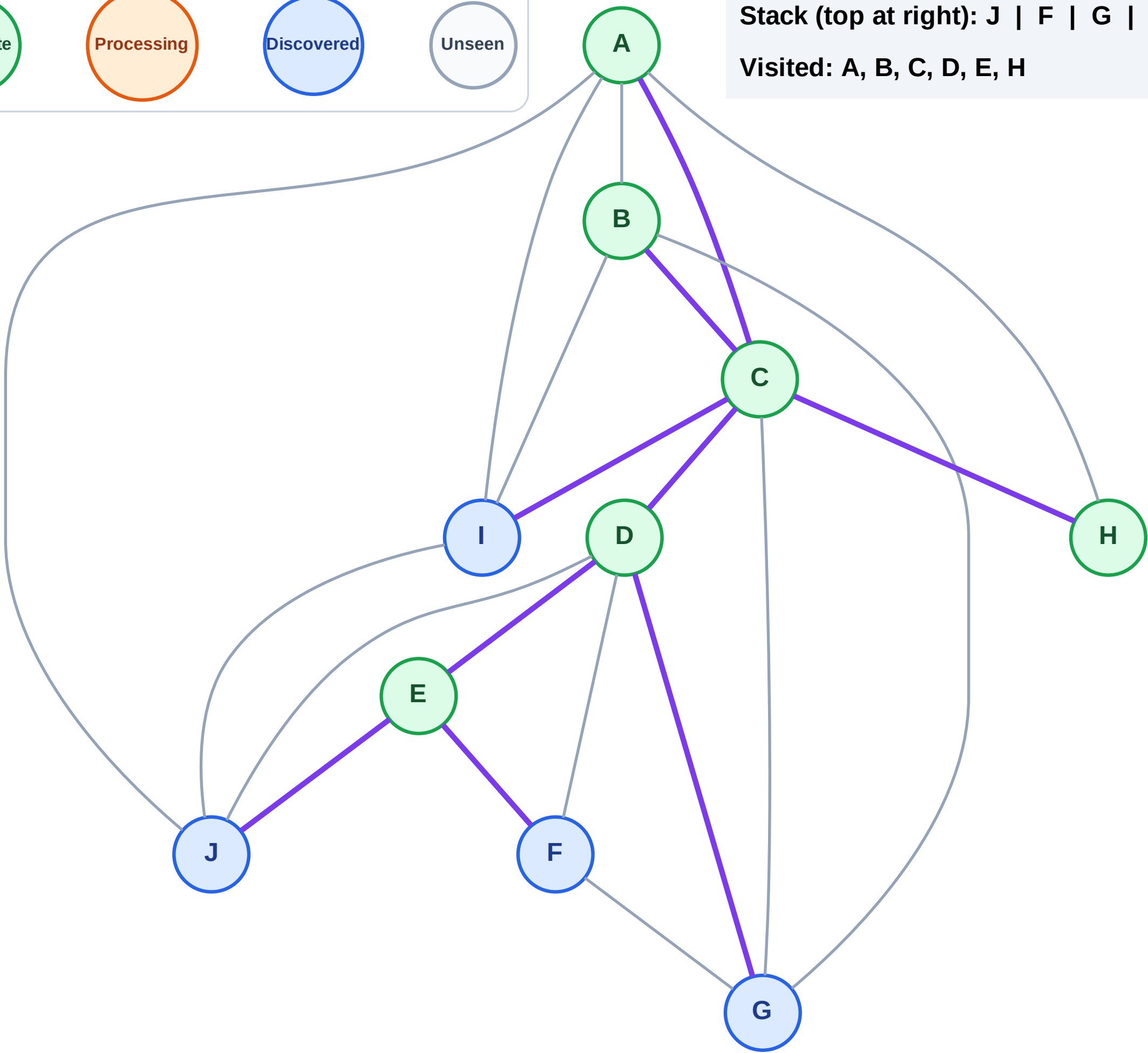
Step 13: No new neighbors from H

Legend



Stack (top at right): J | F | G | I

Visited: A, B, C, D, E, H



DFS Traversal

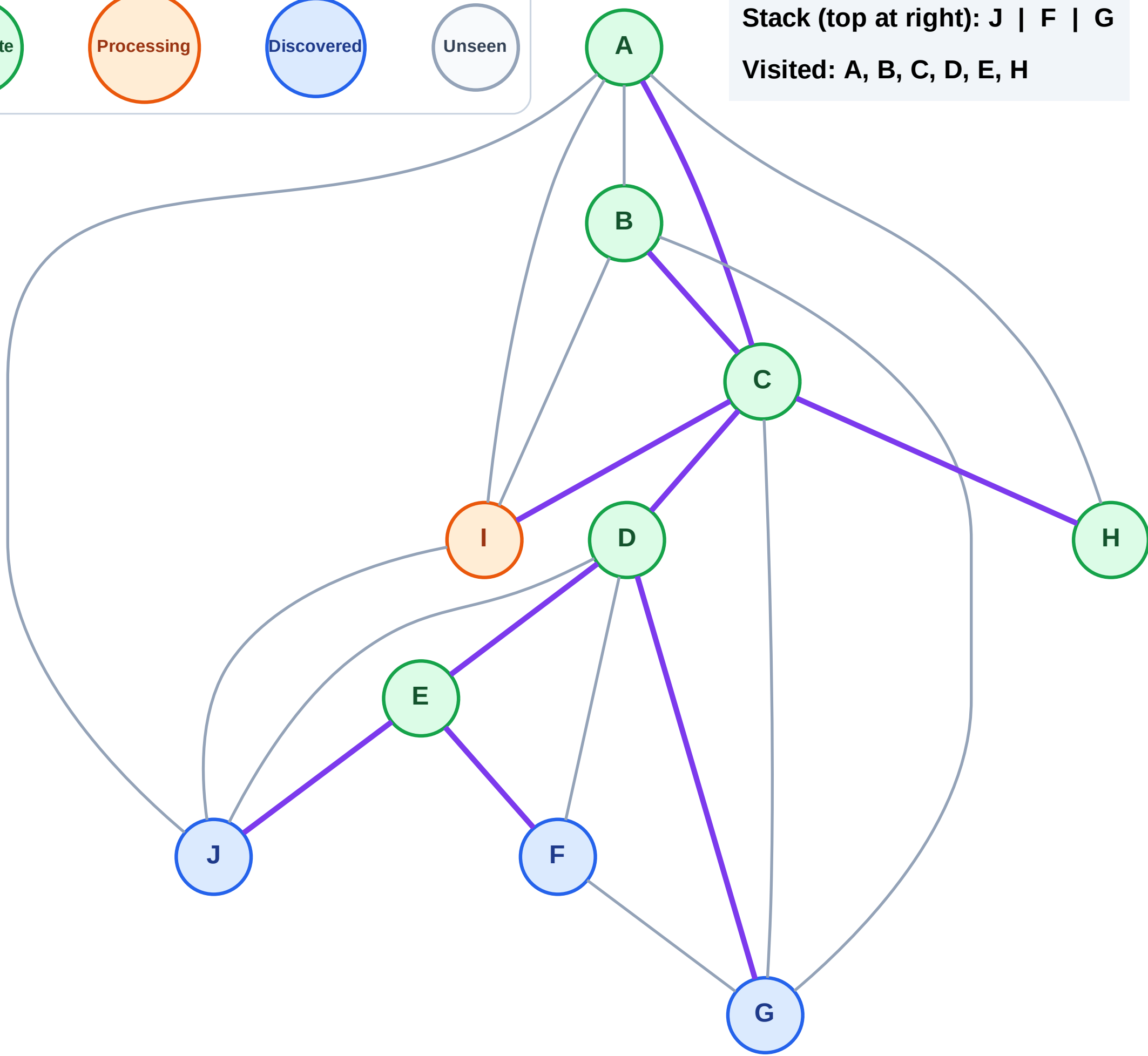
Step 14: Process node I

Legend



Stack (top at right): J | F | G

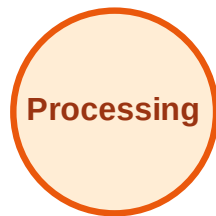
Visited: A, B, C, D, E, H



DFS Traversal

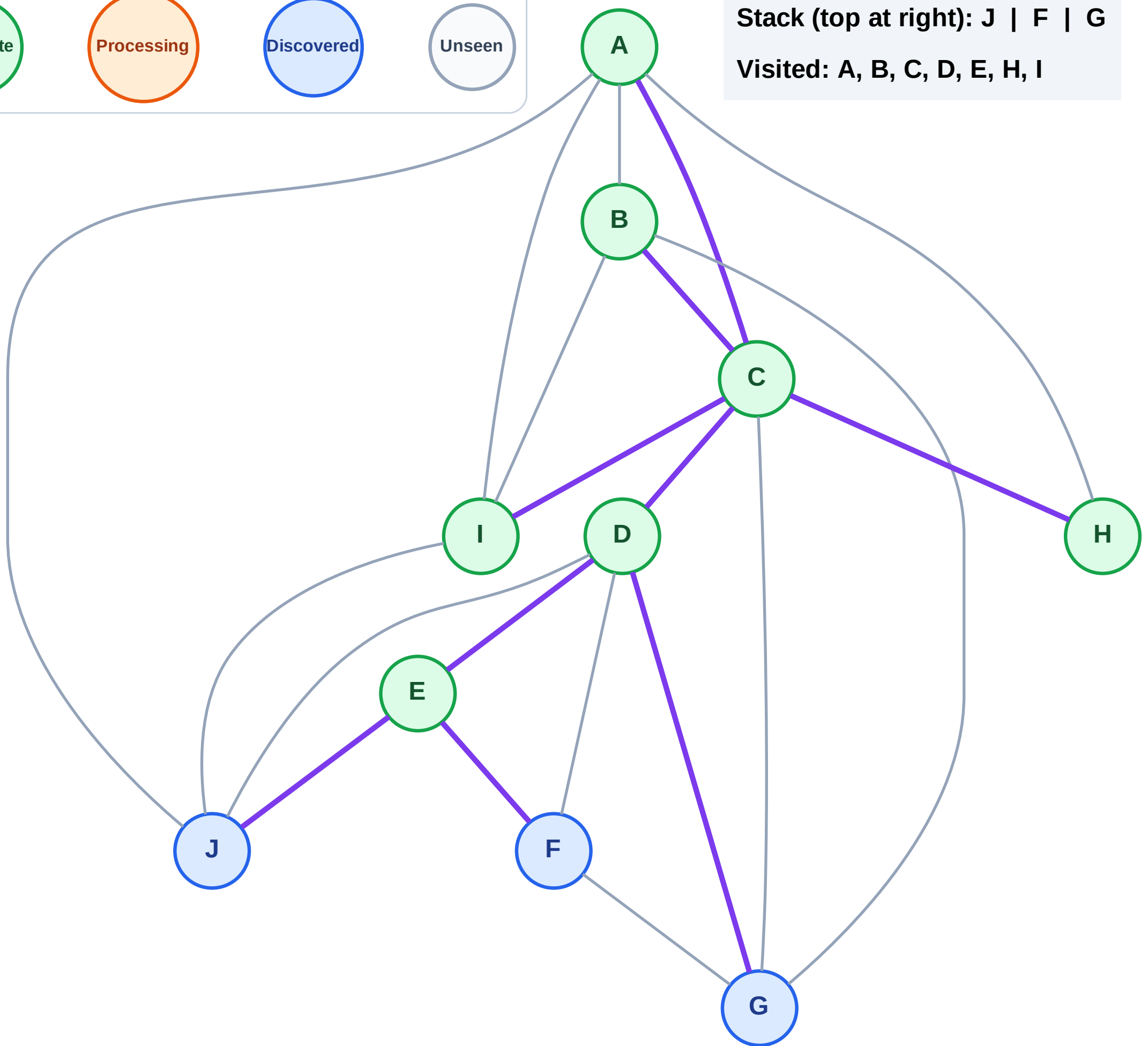
Step 15: No new neighbors from I

Legend



Stack (top at right): J | F | G

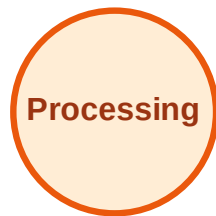
Visited: A, B, C, D, E, H, I



DFS Traversal

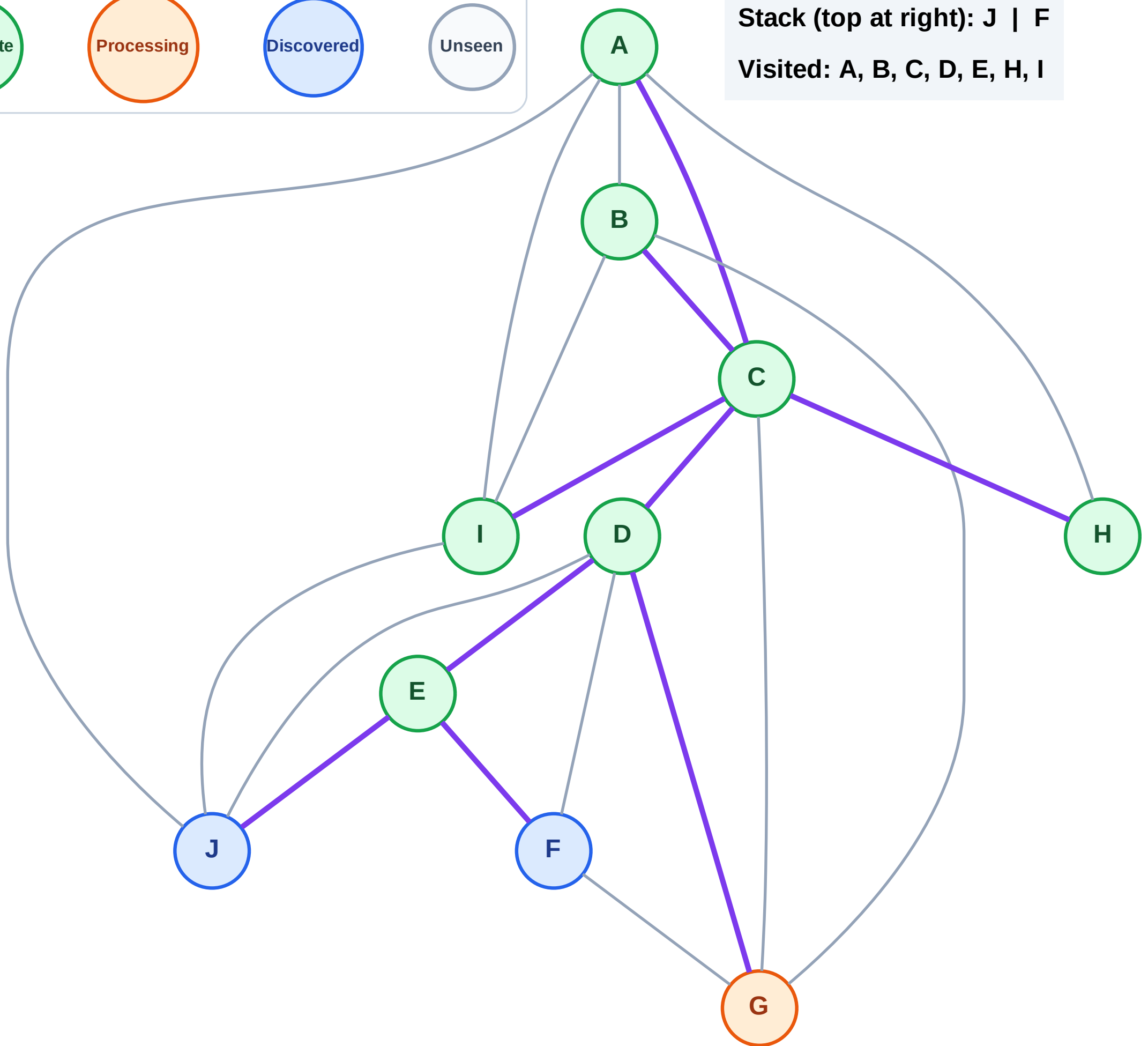
Step 16: Process node G

Legend



Stack (top at right): J | F

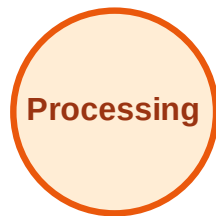
Visited: A, B, C, D, E, H, I



DFS Traversal

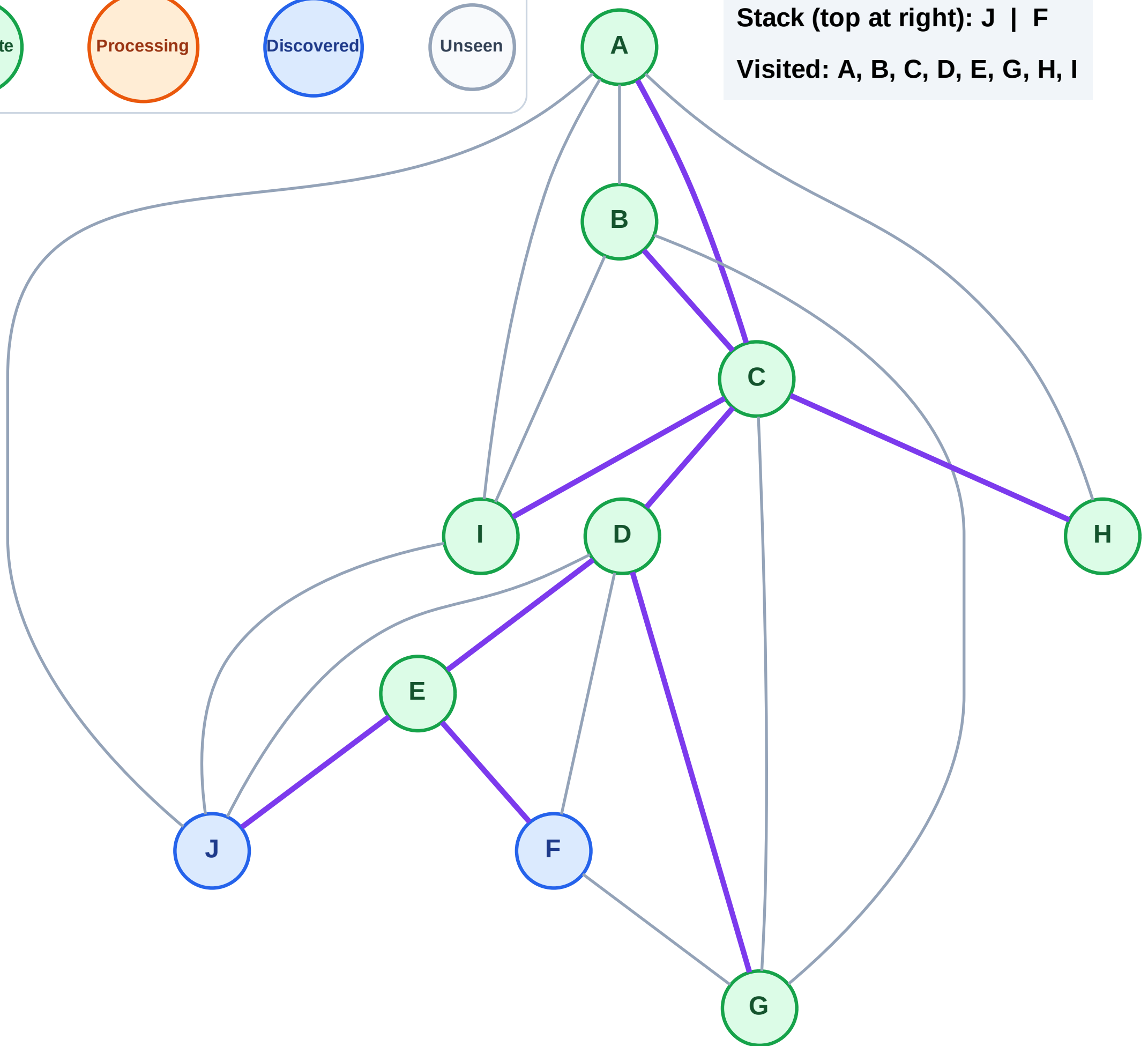
Step 17: No new neighbors from G

Legend



Stack (top at right): J | F

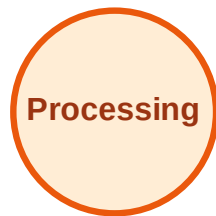
Visited: A, B, C, D, E, G, H, I



DFS Traversal

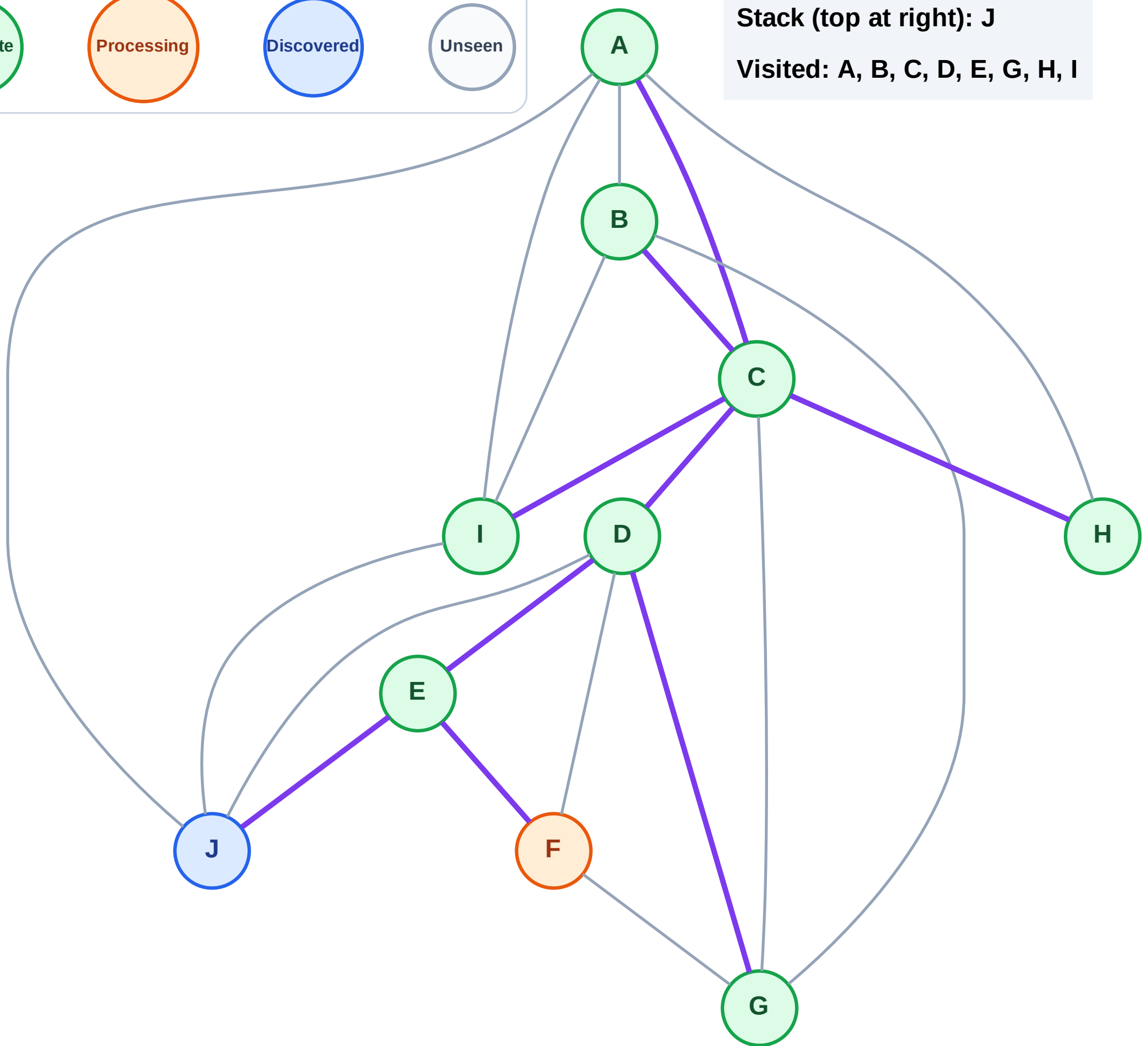
Step 18: Process node F

Legend



Stack (top at right): J

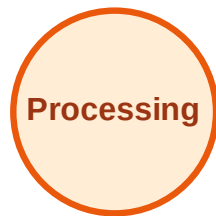
Visited: A, B, C, D, E, G, H, I



DFS Traversal

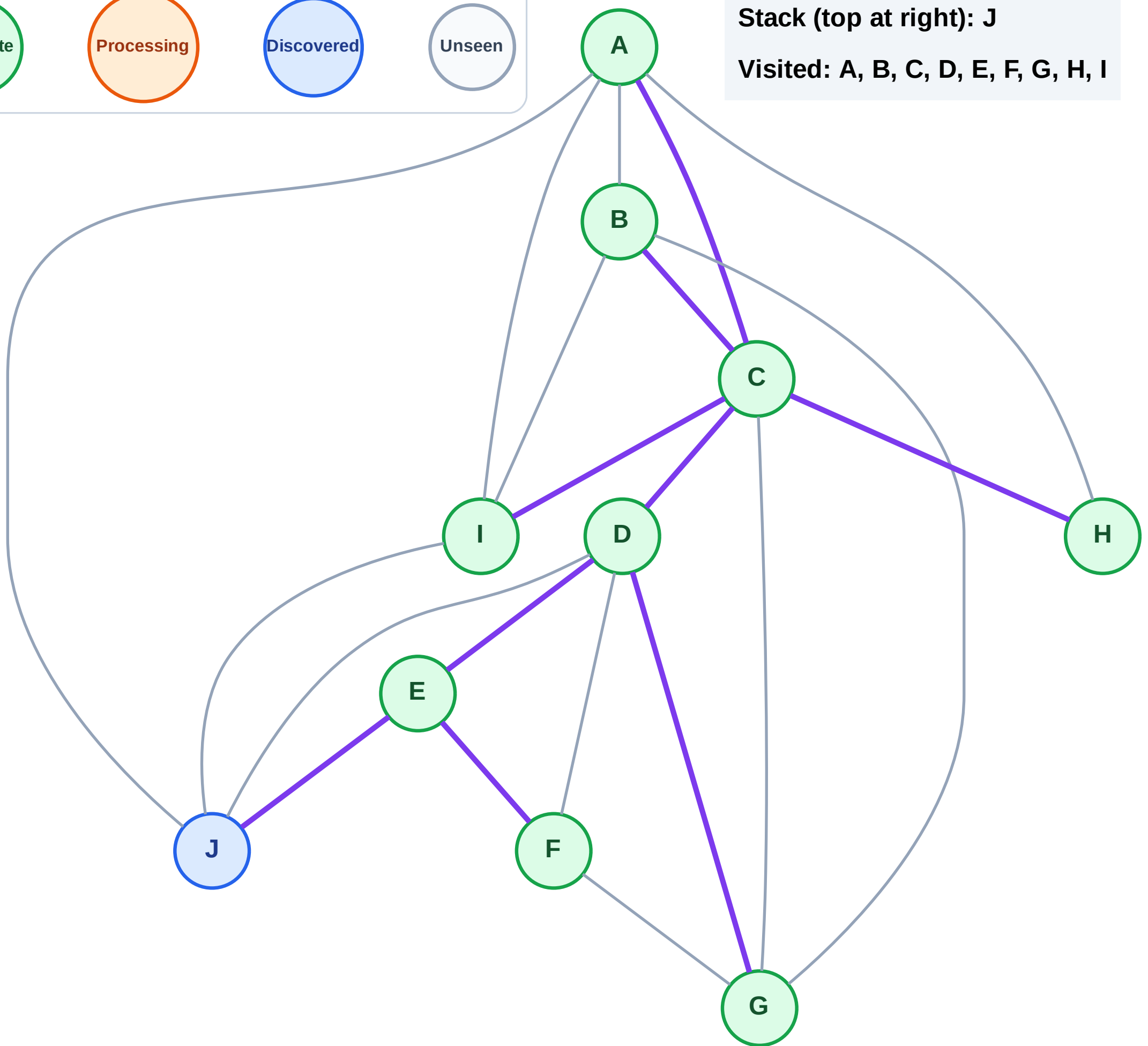
Step 19: No new neighbors from F

Legend



Stack (top at right): J

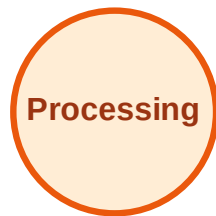
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

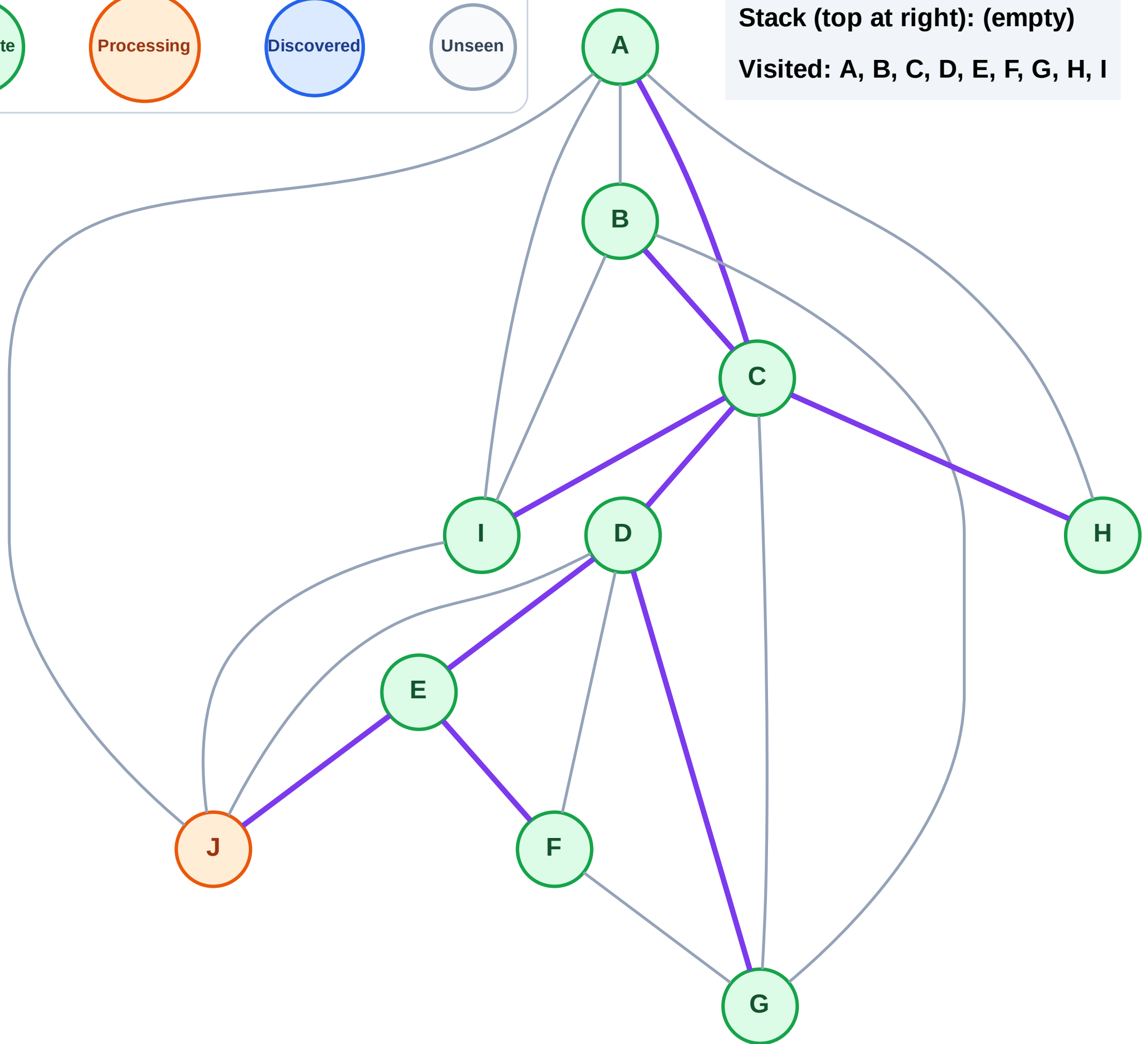
Step 20: Process node J

Legend



Stack (top at right): (empty)

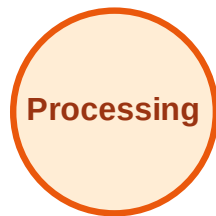
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

